



OPEN Optimized super-twisting active disturbance rejection control based on nonlinear extended state observer for aircraft attitude stabilization

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In this paper, a novel control strategy for the attitude stabilization of a nonlinear aircraft model is proposed, integrating active disturbance rejection control (ADRC) with the smart super-twisting sliding mode control (ST-SMC) algorithm. The primary objective of this study is to enhance the control system's robustness against disturbances while improving tracking accuracy. The proposed method leverages a nonlinear extended state observer (NESO) to accurately estimate the lumped disturbances and system uncertainties. These estimated disturbances are actively compensated within the control law, thereby transforming the uncertain nonlinear system into a simplified compensated system. To ensure robust tracking and effective disturbance rejection, a high-order sliding mode control law based on the super-twisting algorithm is implemented, where the gains of the super-twisting controller are tuned using the reinforcement learning (RL) method. The stability of the nonlinear extended state observer and the super-twisting ADRC is theoretically analyzed, and closed-loop stability is proved. Extensive simulations are conducted on a nonlinear aircraft model under various disturbance scenarios, including actuator faults, aerodynamic uncertainties, and atmospheric disturbances. The results demonstrate the superiority of the proposed method over traditional control strategies, such as Linear ADRC and classical ST-SMC, in terms of convergence speed, robustness, and disturbance rejection performance. This study contributes significantly to flight control systems, offering a robust and efficient solution for modern aerospace applications.

Keywords Nonlinear aircraft model, Active disturbance rejection control (ADRC), Nonlinear extended state observer (NESO), Super-Twisting sliding algorithm (STA), Reinforcement learning (RL)

List of symbols

α	Attack angle (deg)
β	Side slip angle (deg)
θ	Pitch angle (deg)
ϕ	Roll angle (deg)
ψ	Yaw angle (deg)
δ_E	Elevator control surface (deg)
δ_A	Aileron control surface (deg)
δ_R	Rudder control surface (deg)
δ_T	Throttle control surface (deg)
ρ	Density (lbs/ft^3)
q	Dynamic pressure (lbs/ft^2)
V	Total velocity (ft/sec)
m	Total mass (lbs)
S, c	Wing area (ft^2), Mean aerodynamic chord Length (ft)

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g	Gravitational acceleration (lbs/sec^2)
u, v, w	Linear velocities in body frame (ft/sec)
p, q, r	Angular velocities in body frame (deg/sec)
x, y, z	Inertial position
$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	Moment of inertia in body frame (N/ft^2)
$U1$	Trim condition velocity (ft/sec)
C_l, C_m, C_n	Moment coefficients of roll, pitch, yaw
C_D, C_y, C_L	Force coefficients along x axis, y axis, z axis
F_{Ax}, F_{Ay}, F_{Az}	Aerodynamic Force in body frame (N)
l_x, m_y, n_z	Aerodynamic moment in body frame ($N.ft$)
F_{Tx}, F_{Ty}, F_{Tz}	Thrust force in body frame (N)
T	Thrust vector (N)
x_T, y_T, z_T	Thrust vector location from mass center (ft)
F_s, M_s	Aerodynamic Force (N) and moment in stability frame ($N.ft$)

Aircraft flying at high altitudes encounter significant variability in their dynamic parameters, exhibiting heightened sensitivity to flight conditions, physical properties, and atmospheric disturbances. In such environments, achieving precise attitude control becomes essential, necessitating continuous adjustments throughout the flight. Consequently, designing a robust control system capable of compensating for external disturbances and accommodating changes in the system's physical parameters is a fundamental aspect of modern flight control research.

In systems characterized by complex non-linear dynamics, traditional control methods such as proportional, integral, and derivative (PID) controllers and linear quadratic regulator (LQR) controllers often fail to track the desired path under conditions of uncertainty and strong disturbances. To address this challenge, active disturbance rejection control (ADRC) was introduced by Han¹. This proposal by Gao et al.^{2,3} was introduced in the English language. ADRC is an advanced control strategy that estimates the disturbances of the entire system, including external disturbances, uncertainties, and unmodeled dynamics, using an extended state observer (ESO). Due to its robust performance and simplicity, ADRC has found widespread applications in numerous practical scenarios^{4–8}.

ADRC consists of two essential components: an Extended State Observer (ESO) and a suitable control law for disturbance rejection. The ESO⁹ is the core of ADRC because it estimates unmodeled dynamics, uncertainties, and external disturbances considered as total the disturbance. Also, it provides estimations of the system states. The ESO has been widely adopted since its initial introduction by Han⁹ primarily due to two key advantages¹⁰: First, when system dynamics are unknown, the ESO can estimate dynamics and disturbances, while guaranteeing an upper bound on the estimation error that decreases with increasing observer bandwidth. Second, when system dynamics are explicitly defined, the observer's error dynamics exhibit asymptotic stability.

Considering the inherent challenges of flight vehicles, a suitable control strategy to be integrated with the extended state observer can be drawn from the class of robust control techniques. Among the various non-linear robust control methods, sliding mode control has maintained a prominent position over the past decades due to its high robustness and simplicity of design. However, a major drawback of SMC is the chattering phenomenon caused by the switching action near the sliding surface. To mitigate this issue, high-order sliding mode control (HOSMC) techniques have emerged as an effective solution. These methods minimize chattering by concealing the high-frequency switching term within higher derivatives of the sliding variable. The super-twisting sliding mode control (ST-SMC), known as a second-order sliding mode technique significantly, reduces chattering while enhancing efficiency, convergence speed, and robustness compared to conventional methods^{11,12}.

One of the main practical challenges lies in tuning multiple control gains, which have a direct impact on system stability, convergence rate, and disturbance rejection capability¹³. Manual adjustment of these gains is often time-consuming and may even result in suboptimal performance, particularly in complex, multidimensional, and nonlinear flight systems¹⁴. With the rapid progress of machine learning and artificial intelligence, data-driven approaches have been increasingly employed to address this limitation. These methods provide the ability to automatically adapt control parameters to varying operating conditions, thereby reducing reliance on manual trial-and-error tuning and improving overall performance. In particular, reinforcement learning offers a powerful framework for online and adaptive gain adjustment, ensuring robustness and adaptability of flight control systems under highly nonlinear dynamics and uncertain environments.

This paper presents a novel active disturbance rejection control (ADRC) framework for nonlinear model aircraft attitude stabilization, integrating a reinforcement learning tuned super-twisting sliding mode controller with a nonlinear extended state observer. The proposed method enhances robustness and adaptability against actuator dynamics, aerodynamic uncertainties, sensor bias, and external disturbances. Key contributions include: (i) development of a super-twisting ADRC structure with reinforcement learning, (ii) design of a nonlinear ESO for disturbance estimation and compensation, and (iii) implementation of a nonlinear tracking differentiator to improve convergence and reduce oscillations. In this paper, in Sect. “[Mathematical model](#)”, the nonlinear aircraft model and actuators' dynamics are described. Then, in Sect. “[Controller methodology](#)”, the non-linear extended state observer for estimating unknown variables is presented. The non-linear active disturbance rejection controller based on a smart super-twisting algorithm is designed to ensure that the closed-loop state variables match the reference state. In the following, the stability of non-linear ESO and ADRC is theoretically analyzed and proved. In Sect. “[Stability analysis of closed-loop dynamics](#)”, the attitude control of the aircraft is simulated, taking into account the limitations of the actuators and the types of disturbances. In Sect. “[Simulation result](#)”, the summary and conclusions about this work are discussed and, suggestions for future work are presented.

The main objective of the paper is attitude control, control of three rotational DOFs (ϕ, θ, ψ), with the three aerodynamic actuators (aileron, elevator, rudder), i.e., a 3-input/3-output design problem. Indeed, although the full model is 6-DOF, the control objective is attitude regulation (ϕ, θ, ψ), achieved through the three control inputs ($\delta_a, \delta_e, \delta_r$). In other words, the problem is formulated as a three-input/three-output (MIMO) control design, which is standard in aircraft attitude control practice. The main objective of this study is to apply attitude control to the six-degree-of-freedom model. The obtained results confirm the capability of the proposed approach in achieving stable and reliable system performance.

Mathematical model

Six-degree of freedom dynamic model

The simulations in this study are conducted on a nonlinear model of the Learjet 24, an American twin-engine business jet. This aircraft typically accommodates 6 to 8 passengers, with a maximum speed of 851 km/h and a flight range of 2,095 km. It is powered by General Electric CJ610-6 engines, generating approximately 13.1 kN of thrust. A three-view diagram of the Learjet 24 is presented in Fig. 1.

The equations of the aircraft model are presented under the following assumptions:

- The flight vehicle is considered as a rigid body.
- Air density, aircraft mass, gravitational acceleration, and thrust force are assumed constant throughout the flight.
- The flight is considered for the descent phase and general flight conditions.

The equations governing the motion of the aircraft are as follows for simulation, are formulated as follows¹⁶:

a) Angular velocities:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} \frac{\{I_{zz}(L+L_T)+I_{xz}(N+N_T)-[I_{xz}(I_{yy}-I_{xx}-I_{zz})p+[I_{xz}^2+I_{zz}(I_{zz}-I_{yy})]r\}q}{(I_{xx}I_{zz}-I_{xz}^2)} \\ \frac{[(M+M_T)-(I_{xx}-I_{zz})pr-I_{xz}(p^2-r^2)]}{I_{yy}} \\ \frac{\{I_{xz}(L+L_T)+I_{xx}(N+N_T)-[I_{xz}(I_{yy}-I_{xx}-I_{zz})r+[I_{xz}^2+I_{xx}(I_{xx}-I_{yy})]p\}q}{(I_{xx}I_{zz}-I_{xz}^2)} \end{bmatrix} \quad (1)$$

where

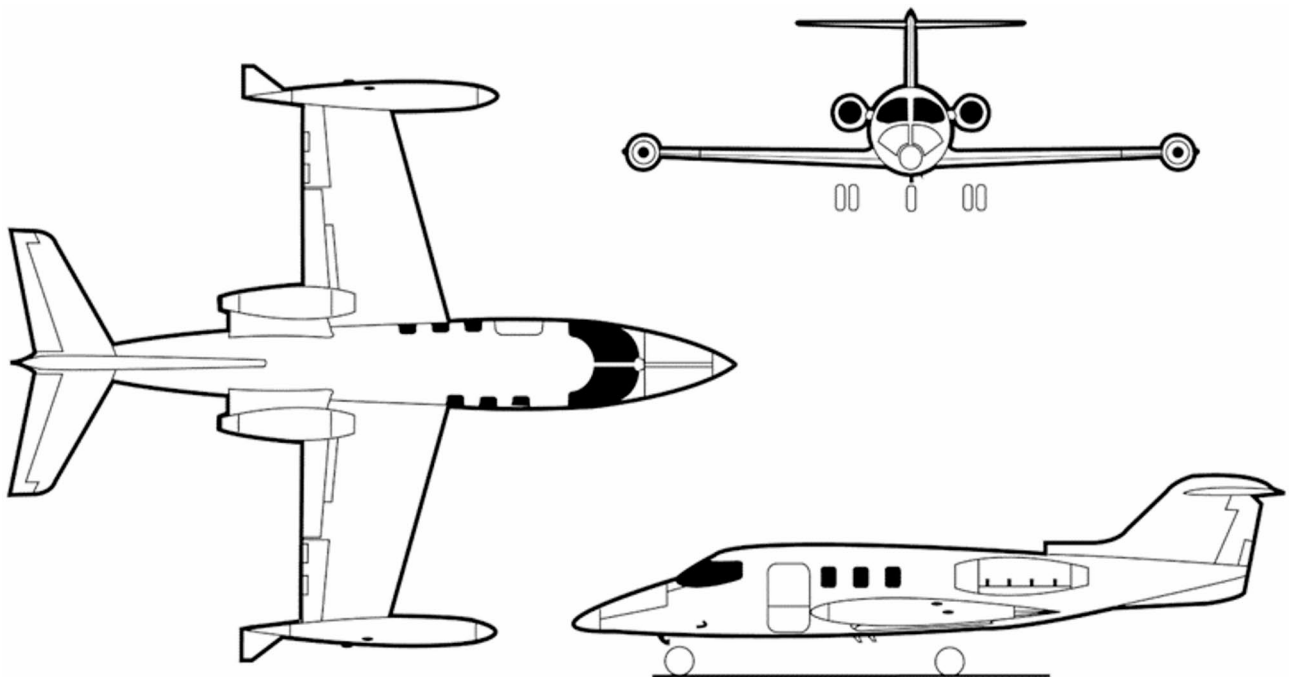


Fig. 1. Learjet 24¹⁵.

$$\begin{bmatrix} L \\ M \\ N \end{bmatrix} = \begin{bmatrix} C_l \bar{q} S b \\ C_m \bar{q} S \bar{C} \\ C_n \bar{q} S b \end{bmatrix} \quad (1-a)$$

$$\begin{aligned} C_l &= (C_{l_0} + C_{l_\beta} \beta + C_{l_{\delta_a}} \delta_a + C_{l_{\delta_r}} \delta_r + C_{l_p} \left(\frac{pb}{2V}\right) + C_{l_r} \left(\frac{rb}{2V}\right)) \\ C_m &= (C_{m_0} + C_{m_\alpha} + C_{m_{\delta_e}} \delta_e + C_{m_u} \left(\frac{V-U_1}{V}\right) + C_{m_q} \left(\frac{q\bar{C}}{2V}\right) + C_{m_{\dot{\alpha}}} \left(\frac{\dot{\alpha}\bar{C}}{2V}\right)) \\ C_n &= (C_{n_0} + C_{n_\beta} \beta + C_{n_{\delta_a}} \delta_a + C_{n_{\delta_r}} \delta_r + C_{n_p} \left(\frac{pb}{2V}\right) + C_{n_r} \left(\frac{rb}{2V}\right)) \end{aligned} \quad (1-b)$$

b) Linear velocities:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} g_x + \frac{1}{m} (F_{A_x} + F_{T_x}) + vr - wq \\ g_y + \frac{1}{m} (F_{A_y} + F_{T_y}) + wp - ur \\ g_z + \frac{1}{m} (F_{A_z} + F_{T_z}) + uq - vp \end{bmatrix} \quad (2)$$

where

$$\begin{bmatrix} F_{A_x} \\ F_{A_y} \\ F_{A_z} \end{bmatrix} = \begin{bmatrix} -C_D \bar{q} S \\ C_y \bar{q} S \\ -C_L \bar{q} S \end{bmatrix} \quad (2-a)$$

$$\begin{aligned} C_D &= (C_{D_0} + C_{D_\alpha} + C_{D_{\delta_e}} \delta_e + C_{D_u} \left(\frac{V-U_1}{V}\right)) \\ C_y &= (C_{y_0} + C_{y_\beta} \beta + C_{y_{\delta_a}} \delta_a + C_{y_{\delta_r}} \delta_r + C_{y_p} \left(\frac{pb}{2V}\right) + C_{y_r} \left(\frac{rb}{2V}\right)) \\ C_L &= (C_{L_0} + C_{L_\alpha} + C_{L_{\delta_e}} \delta_e + C_{L_u} \left(\frac{V-U_1}{V}\right) + C_{L_q} \left(\frac{q\bar{C}}{2V}\right) + C_{L_{\dot{\alpha}}} \left(\frac{\dot{\alpha}\bar{C}}{2V}\right)) \end{aligned} \quad (2-b)$$

c) Euler angles:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} p + q \sin \phi \tan \theta + r \cos \phi \tan \theta \\ q \cos \phi - r \sin \phi \\ (q \sin \phi + r \cos \phi) \sec \theta \end{bmatrix} \quad (3)$$

d) Position:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \cos \theta \cos \psi & \sin \phi \sin \theta \cos \psi - \cos \phi \sin \psi & \cos \phi \sin \theta \cos \psi - \sin \phi \sin \psi \\ \cos \phi \sin \psi & \sin \phi \sin \theta \sin \psi + \cos \phi \cos \psi & \cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi \\ -\sin \theta & \sin \phi \cos \theta & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad (4)$$

e) Gravitational acceleration matrix components in the body frame are:

$$\begin{bmatrix} g_x \\ g_y \\ g_z \end{bmatrix} = \begin{bmatrix} -g \sin \theta \\ g \sin \phi \cos \theta \\ g \cos \phi \cos \theta \end{bmatrix} \quad (5)$$

f) Other required equations to access other flight parameters are:

$$\begin{aligned} \alpha &= \tan^{-1} \left(\frac{w}{u} \right) \\ \beta &= \sin^{-1} \left(\frac{v}{V} \right) \\ V &= \sqrt{u^2 + v^2 + w^2} \\ \dot{\alpha} &= \frac{\dot{w}u - u\dot{w}}{u^2 + w^2} \\ \bar{q} &= \frac{1}{2} \rho V^2 \end{aligned} \quad (6)$$

• δ_T is the angle of engine installation, thrust vector in the body frame is:

$$\begin{bmatrix} F_{T_x} \\ F_{T_y} \\ F_{T_z} \end{bmatrix} = \begin{bmatrix} T \cos \delta_T \\ 0 \\ -T \sin \delta_T \end{bmatrix} \quad (7)$$

h) Engine Model:

The torque of each engine is calculated according to the location of the engine and is added to the torque of the aircraft.

$$M_{Engine} = \begin{bmatrix} L_T \\ M_T \\ N_T \end{bmatrix} = R \times T \quad (8)$$

$$R = \begin{bmatrix} x_{Engine} \\ y_{Engine} \\ z_{Engine} \end{bmatrix} \quad (8-a)$$

Ultimately, a set of twelve equations corresponding to the six-degree-of-freedom model is derived and solved using the prescribed initial conditions. The governing equations for forces and torques in the wind frame are established, and the associated transformation matrices for converting these quantities into the body frame are provided in Appendix A.

Actuators dynamics

The dynamics of control surface actuators, including those for the aileron, elevator, and rudder, are commonly modeled as second-order dynamic systems. The general form of the transfer function is given by:

$$\begin{aligned} G_{\text{aileron-actuator}}(s) &= \frac{\delta_a(s)}{\delta_{a-command}(s)} = \frac{\omega_a^2}{s^2 + 2\zeta_a \omega_a s + \omega_a^2} \\ G_{\text{elevator-actuator}}(s) &= \frac{\delta_e(s)}{\delta_{e-command}(s)} = \frac{\omega_e^2}{s^2 + 2\zeta_e \omega_e s + \omega_e^2} \\ G_{\text{rudder-actuator}}(s) &= \frac{\delta_r(s)}{\delta_{r-command}(s)} = \frac{\omega_r^2}{s^2 + 2\zeta_r \omega_r s + \omega_r^2} \end{aligned} \quad (9)$$

Where $\delta_i(s)$, $i = a, e, r$ is the actual surface deflection (output), $\delta_{i-command}(s)$, $i = a, e, r$ is the commanded surface deflection (input), ω_i ($\frac{rad}{s}$), $i = a, e, r$ is the natural frequency of the actuator, and ζ_i , $i = a, e, r$ is the damping ratio. Tuning the parameters of actuator dynamics is crucial for achieving optimal performance and stability of the aircraft control system. It depends on the physical limitations of the actuators, control surfaces, flight conditions, and aircraft design requirements.

Controller methodology

If the dynamic system is considered as follows:

$$\ddot{\mathbf{y}}(t) = \mathbf{F}(\mathbf{y}(t), \mathbf{u}(t), \mathbf{d}(t)) + \mathbf{B}\mathbf{u}(t) \quad (10)$$

Where $\mathbf{y}(t)$ is the output of the system, $\mathbf{u}(t)$ is the control input signal, $\mathbf{B}$ is the gain of the control signal, $\mathbf{d}(t)$ is an external disturbance, $\mathbf{F}$ is considered as the total disturbance, which is a multivariate function of the internal dynamics of the system and the external disturbance.

If $\mathbf{x}_1 = \mathbf{y}$, $\mathbf{x}_2 = \dot{\mathbf{y}}$, and $\mathbf{x}_3 = \mathbf{F}$ are considered. Assuming that $\mathbf{F}$ is differentiable and $\mathbf{H} = \dot{\mathbf{F}}$, in this case, Eq. (10) can be expressed in the form of state space:

$$\begin{cases} \dot{\mathbf{x}}_1 = \mathbf{x}_2 \\ \dot{\mathbf{x}}_2 = \mathbf{x}_3 + \mathbf{B}\mathbf{u}(t) \\ \dot{\mathbf{x}}_3 = \mathbf{H}(t) \\ \mathbf{y} = \mathbf{x}_1 \end{cases} \quad (11)$$

Where $\mathbf{x}_1 = [\phi, \theta, \psi]^T$ is the 3×1 vector of Euler angles, $\mathbf{x}_2 = [p, q, r]^T$ is the 3×1 vector of Euler rate angles, $\mathbf{x}_3 = [x_{31}(t), x_{32}(t), x_{33}(t)]^T$ is the 3×1 vector of total disturbance, $\mathbf{H}(t) = [H_1(t), H_2(t), H_3(t)]^T$ is the 3×1 vector of finite and differentiable functions, $\mathbf{u}(t) = [\delta_a, \delta_e, \delta_r]^T$ is the 3×1 vector of the control input signals, $\mathbf{B}$ is a 3×3 matrix of control signal gains, and $\mathbf{y} = [\phi, \theta, \psi]^T$ is the 3×1 vector of system outputs.

Tracking differentiator

The tracking differentiator is designed to generate smooth signals while maintaining rapid response characteristics. When the initial condition of the tracking differentiator matches that of the system, it significantly reduces overshoot. The tracking differentiator consists of a nonlinear element and a transient profile generator, ensuring that the system responds smoothly without abrupt variations.

A modified tracking differentiator is proposed as follows:

$$\begin{cases} \dot{v}_1 = v_2 \\ \dot{v}_2 = -r \operatorname{sat}(v_1 - \mu(t) + \frac{v_2 |v_2|}{2r}) \end{cases} \quad (12)$$

Where $\mu(t)$ is the reference signal, v_1 is the filtered tracking signal, v_2 is the derivative of the filtered signal, and r is used to adjust the convergence speed and accuracy. To illustrate the effectiveness of tracking accuracy, a sinusoidal signal containing white noise has been processed using the tracking differentiator, and the result is presented in Fig. 2.

Figures 2 and 3 confirm that TD effectively filters noise while maintaining accurate differentiation. However, proper tuning of TD parameters is essential to achieve a balance between responsiveness and noise suppression.

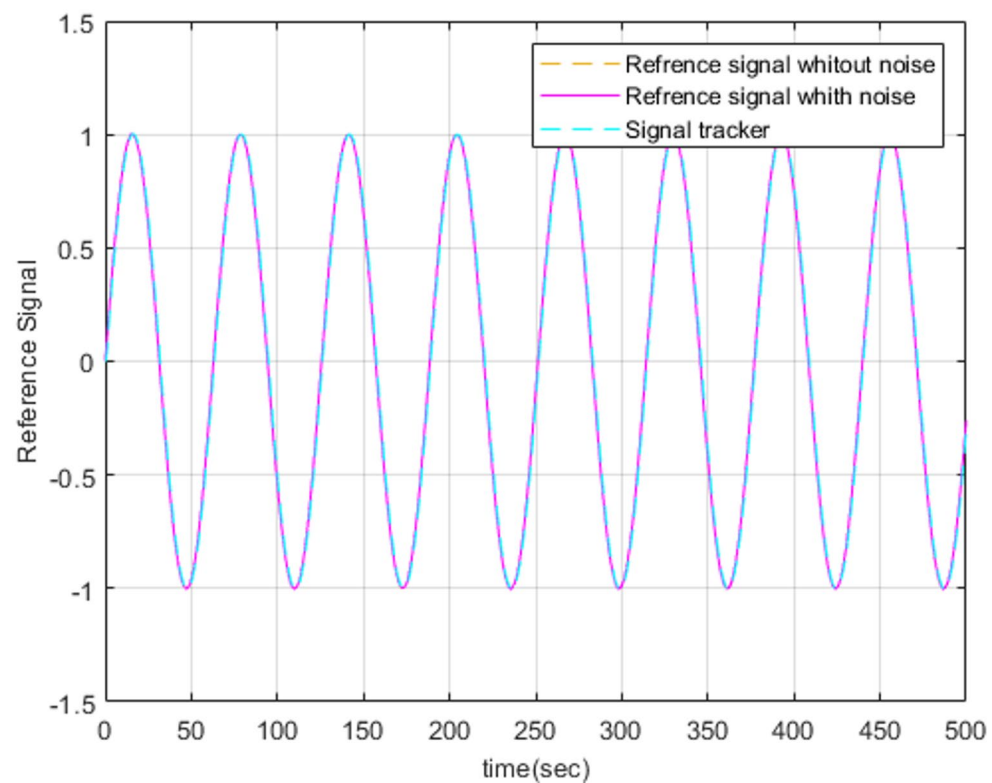


Fig. 2. Tracking of a noisy sinusoidal signal by TD.

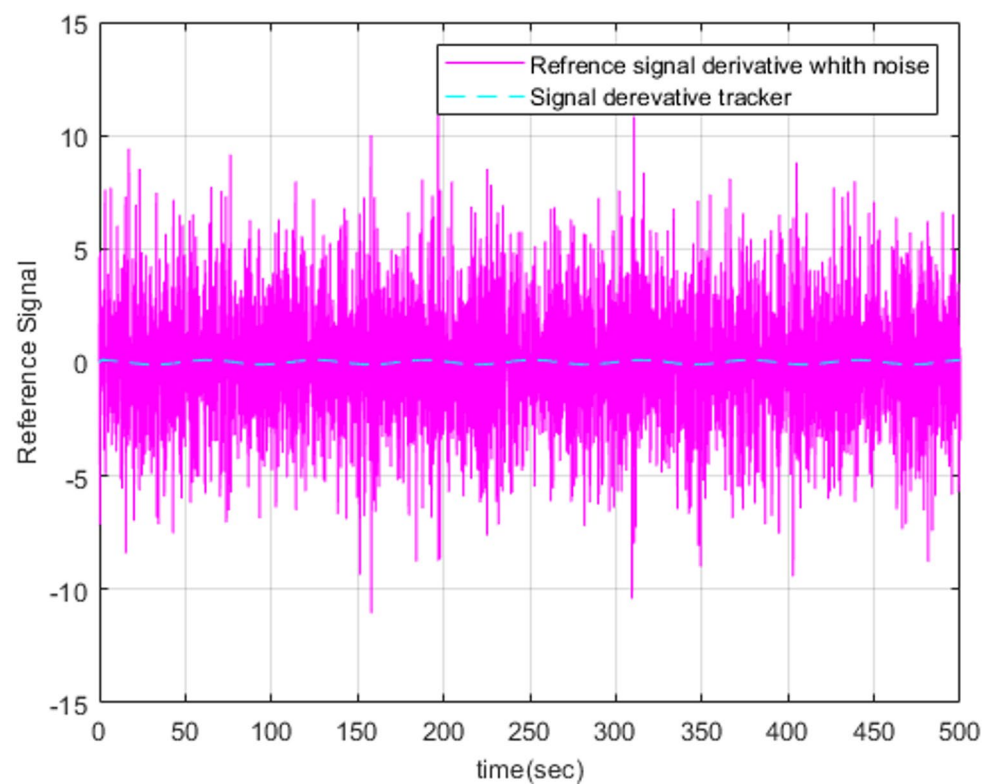


Fig. 3. Derivative tracking of a noisy sinusoidal signal by TD over time.

Non-linear extended state observer

In real-world industrial systems, uncertainties and external disturbances can significantly degrade system performance. Traditional control methods, including classical and optimal control, heavily depend on precise system models, rendering them less effective in the presence of uncertainties and unmodeled dynamics. This limitation has led to the development of robust control strategies. In an ESO-based robust control framework, the system model, measurable states, external disturbances, and uncertainties are estimated in real-time to improve performance.

The non-linear extended state observer for Eq. (11) is formulated as follows:

$$\begin{cases} e = z_1 - x_1 \\ \dot{z}_1 = z_2 - l_1 e \\ \dot{z}_2 = z_3 + b_0 u(t) - l_2 e \\ \dot{z}_3 = -l_3 \text{fal}(e, \sigma, \delta) \end{cases} \quad (13)$$

Where $z_1 = [\hat{\phi}, \hat{\theta}, \hat{\psi}]^T$ is the estimation vector of x_1 , $z_2 = [\hat{p}, \hat{q}, \hat{r}]^T$ is the estimation vector of x_2 , $z_3 = [\hat{H}_1(t), \hat{H}_2(t), \hat{H}_3(t)]^T$ is the extended vector used to estimate the total disturbance, $e = z_1 - x_1$

is the 3×1 vector of the angle estimation error, $l_i, i = 1, 2, 3$ is the 3×3 matrix of observer gains, and b_0 is the 3×3 matrix of control input gains.

simplify the observer gains tuning process, Zhao proposed a single-parameter tuning approach, where all observer poles are positioned at the same location. This approach ensures that the observer's performance is governed by a single tuning parameter, thereby significantly reducing design complexity. The characteristic equation of the observer is aligned with the desired characteristic equation, ensuring that all roots are equal to $-\omega_0^3$. Then:

$$s^3 + l_1 s^2 + l_2 s + l_3 = (s + \omega_0)^3 \quad (14)$$

Where ω_0 is the bandwidth of the ESO and:

$$l_1 = 3\omega_0, l_2 = 3\omega_0^2, l_3 = \omega_0^3 \quad (15)$$

The selection of ω_0 is critical and should be guided by the system dynamics and performance requirements. Generally, a higher value of ω_0 value results in faster estimation but may increase sensitivity to noise, while a lower ω_0 value leads to slower but more stable estimation.

The non-linear function $\text{fal}(\cdot)$ is describe as follow:

$$\text{fal}(e, \sigma, \delta) = \begin{cases} e^{\frac{\sigma}{\delta}} & |e| \leq \delta \\ \delta \text{sign}(e) & |e| \geq \delta \end{cases} \quad (16)$$

Where $0 < \sigma < 1, \delta > 0$.

The non-linear function $\text{fal}(\cdot)$ is introduced to enhance the observer's performance in the presence of noise and disturbances. The parameters σ and δ in the $\text{fal}(\cdot)$ function provide additional flexibility in tuning the observer's response to different magnitudes of estimation error. Proper selection of these gains ensures that the NESO can effectively estimate both the system states and the lumped disturbances, contributing to the overall robustness of the control system.

Super twisting algorithm

Sliding mode control (SMC), also known as variable structure control, is a robust control methodology characterized by high-speed switching control laws. This technique directs system state variables toward a predefined sliding surface within the state space. The SMC control law consists of two distinct components: one that guides the system states to the sliding surface (reaching mode) and another that keeps them on this surface (sliding mode).

SMC offers significant advantages, including robustness against external disturbances and uncertainties, and rapid transient response. However, a major drawback of conventional SMC is the chattering phenomenon, which arises from the high-frequency switching of control signals. To address this issue, higher-order sliding mode control (HOSMC) has been developed as an effective solution. HOSMC represents a powerful alternative to classical SMC, leveraging the same measurement information to mitigate or eliminate chattering while providing enhanced robustness and control efficiency in the presence of uncertainties and external disturbances¹⁷. Among the most prominent HOSMC algorithms is the super-twisting control law, which combines a discontinuous time derivative term with a continuous function of the sliding variable. The trajectories produced by the super-twisting algorithm are characterized by their rotational motion around the origin in the sliding variable's phase portrait, as shown in Fig. 4.

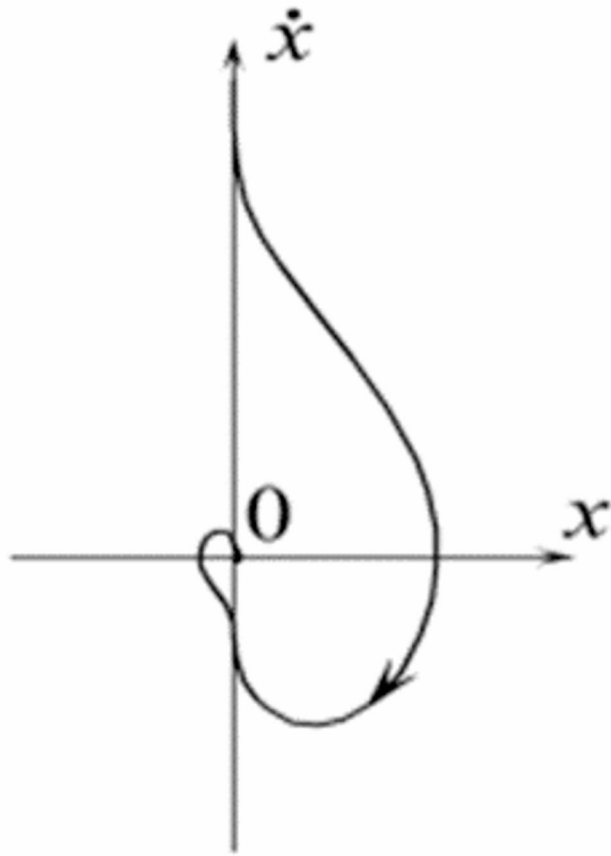


Fig. 4. Super-twisting controller phase portrait.

If $x_1 = [\phi, \theta, \psi]^T$ is the actual state and x_1^d is the 3×1 desired trajectory vector. It is possible to bring the actual values to the desired values by designing a proper control method. For this purpose, the tracking error is defined as follows:

$$\varepsilon = x_1 - x_1^d, \dot{\varepsilon} = x_2 - x_2^d = \dot{x}_1 - \dot{x}_1^d \quad (17)$$

ε and $\dot{\varepsilon}$ are 3×1 vectors representing the error and its derivative, respectively. With assumptions:

- 1) All system states can be measured or estimated.
- 2) The desired values and their first derivative are bounded.
- 3) External disturbances and their first derivative are bounded, satisfying the following inequality:

$$|\dot{d}| \leq L, |d| \leq l = L \quad (18)$$

The sliding surface is defined as follows:

$$s = \dot{\varepsilon} + \lambda \varepsilon \quad (19)$$

Where s is the 3×1 vector and λ is a positive definite 3×3 matrix. The super-twisting control law is defined as follows¹⁸:

$$u_0 = \ddot{x}_1^d - \lambda \dot{\varepsilon} - k_1 |s|^{\frac{1}{2}} \text{sign}(s) - k_2 \int_0^t \text{sign}(s(\tau)) d\tau \quad (20)$$

u_0 is the 3×1 vector and k_1, k_2 are 3×3 positive definite matrix of the super-twisting algorithm gains. Considering Eq (18), the following conditions are imposed on the gains to guarantee stability of the closed-loop system:

$$k_1^2 > 4k_2, k_2 > LI_3 \quad (21)$$

Controller gain tuning via RL-DDPG

As a subset of machine learning, deep reinforcement learning (DRL) has been successfully applied across various domains. As an unsupervised learning approach, DRL is capable of establishing a direct mapping between observations and desired outputs using the maximum likelihood method. Through continuous interaction with the environment guided by policy strategies and a reward function it can iteratively discover and refine this mapping¹⁹. Two main classical DRL algorithms are the Deep Q-Network (DQN)²⁰ and Deep Deterministic Policy Gradient (DDPG)²¹. While DQN is limited to discrete state systems, such as chess, DDPG extends it to handle continuous states and actions. Later improvements, such as Twin Delayed DDPG (TD3) and Asynchronous Advantage Actor-Critic (A3C), have been developed to address specific shortcomings of the original methods, though they introduce their own limitations.

In this work, DDPG is employed to determine the STA-ADRC controller parameters. The essence of DDPG lies in obtaining the optimal policy through continuous interaction between the environment and the agent. Figure 5. presents a block diagram based on a Deep Reinforcement Learning (DRL)-based optimization process where the DRL agent continuously interacts with an environment by receiving the system's state and reward, and sending back actions. This loop continues until an optimization convergence condition is checked. If the condition is not met, the agent keeps interacting and learning; if it is met, the process outputs the optimized parameters.

More specifically, DDPG can be formulated within the framework of a Markov Decision Process (MDP) (S, A, P, R) where S denotes the set of states, A the set of actions, R the reward function. The state transition probability P is expressed as:

$$p(s' | s, a) = p[s_{t+1} = s' | s_t = s, a_t = a] \quad (22)$$

The policy $\pi(a|s) = p[a_t = a | s_t = s]$ specifies the probability of selecting action a given state s . For a trajectory $\tau = \{s_1, a_1, s_2, a_2, \dots, s_T\}$ the probability is given by:

$$p(\tau) = p(s_1) \prod_{t=1}^{T-1} \pi(a_t | s_t) p(s_{t+1} | s_t, a_t) \quad (23)$$

If the cumulative return along τ is $R_c(\tau)$, the expected return under policy π becomes:

$$J(\pi) = \sum_{\tau} R_c(\tau) p(\tau) = E_{\tau \sim \pi} [R_c(\tau)] \quad (24)$$

The objective is to find $\pi^* = \operatorname{argmax}_{\pi} J(\pi)$.

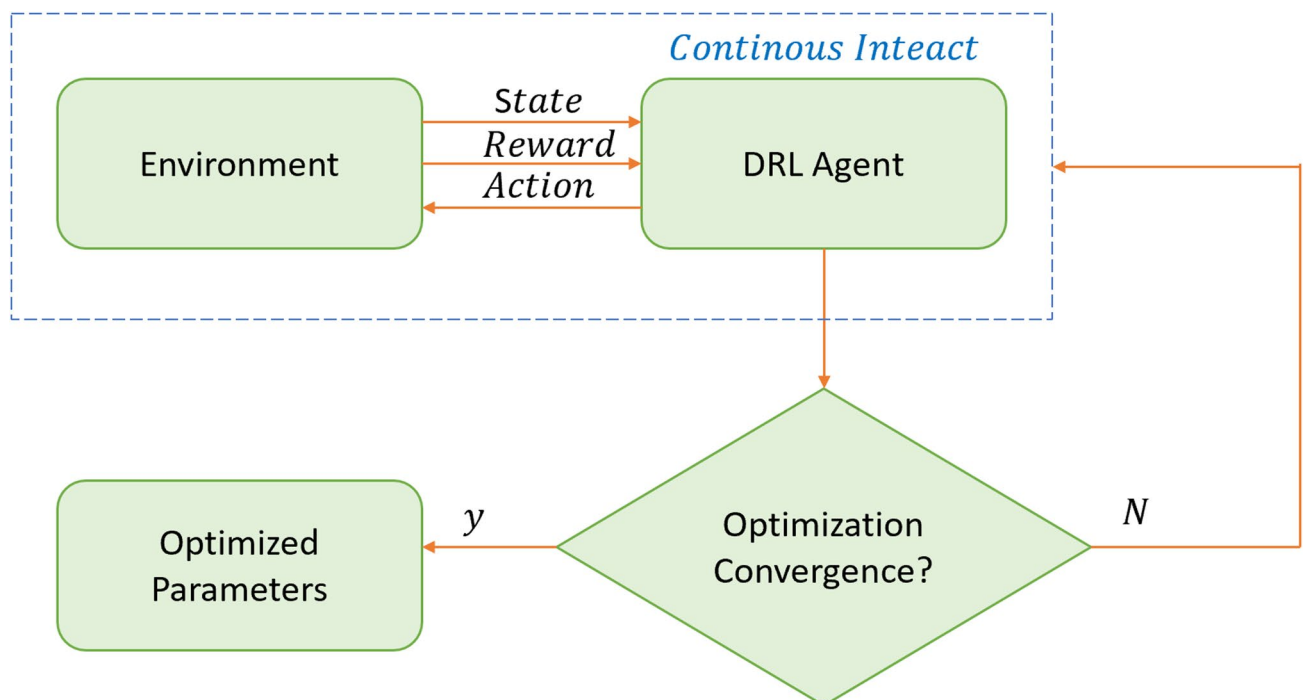


Fig. 5. DRL parameters optimization method.

Description	Value	Description	Value
Number of Actor neurons	64	Critic activation function (1 = ReLU, 2 = tanh, 3 = sigmoid)	ReLU
Number of Actor layers	2	Discount factor	0.99
Actor activation function (1 = ReLU, 2 = tanh, 3 = sigmoid)	ReLU	Mini-batch size	128
Number of Critic neurons	128	Experience buffer length	1e-6
Number of Critic layers	2	Target smoothing factor	1e-3
Learning rate	1e3	Target update frequency	2

Table 1. Reinforcement learning parameters.

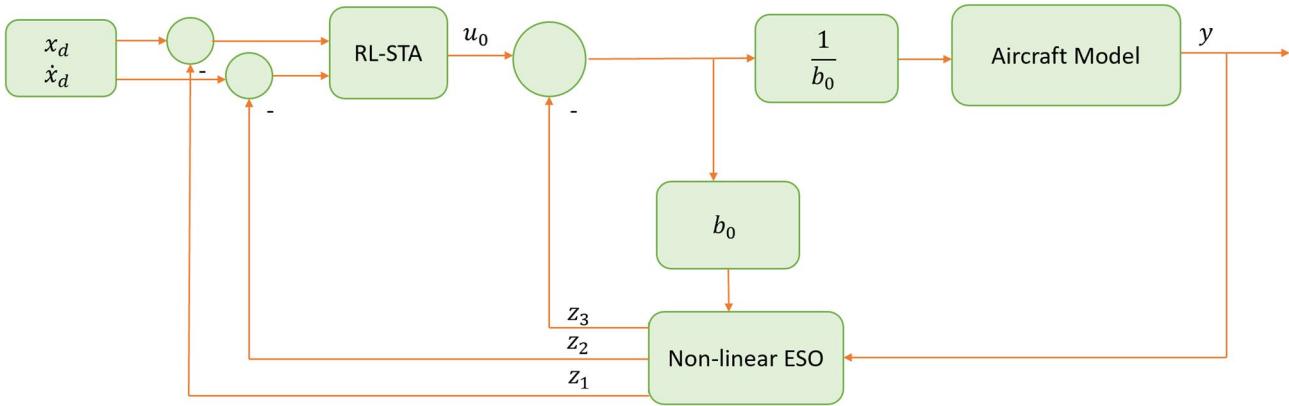


Fig. 6. ADRC Topology.

In DDPG, the agent is composed of four neural networks two primary and two target networks and a replay buffer that stores experience tuples (s_t, a_t, r_t, s_{t+1}) for training. The main and target networks share architecture but differ in weight parameters. The Critic Network takes both the state and the action proposed by the actor network as input and outputs the estimated Q -value:

$$Q(s, a) = E[R_c | s_t = s, a_t = a] \tag{25}$$

The actor network maps the current state directly to a continuous action.

In this study, the reinforcement learning framework is implemented using a Simulink-based simulation environment, in which the nonlinear aircraft model served as the training environment. The system state vector provided to the DRL agent comprised the Euler angles $[\phi, \theta, \psi]^T$, their angular rates $[p, q, r]^T$, and the control input signals $[\delta_a, \delta_e, \delta_r]^T$ applied to the aircraft. The action vector $a(t) = [\lambda, k_1, k_2]$ is defined as the nine tunable gain parameters of the STA-ADRC controller, which are continuously adjusted by the agent during the training process.

The reward function is designed based on the tracking error of the controller and its time derivative $[e, \dot{e}]$, encouraging the agent to minimize both steady-state error and transient deviations. Specifically, a penalty term is included for large error derivatives to suppress oscillatory responses. A termination criterion is imposed by defining a threshold on the absolute tracking error; if the error remains below this value for a specified duration, the episode is terminated early to accelerate convergence.

Training is conducted using the DDPG algorithm in a continuous action space, enabling smooth parameter updates. To ensure closed-loop stability during training, the output gains from the agent are passed through a saturation function, constraining them to a predefined positive range before application to the controller. This approach allowed the RL agent to autonomously discover optimal gain values for the nine controller parameters, enhancing robustness and reducing dependency on manual tuning. The reinforcement learning parameters are presented in Table 1.

ADRC structure

The structure of the active disturbance rejection controller based on the super-twisting algorithm is presented in Fig. 6.

According to Fig. 6, first, the reference values are entered into the tracking differentiator, the output of which is a signal without fluctuation and its differential. Then the output of the tracking accuracy and the output of the non-linear extended state observer are compared and the error and the error differential are formed. The error and the error differential are entered into the super-twisting sliding mode controller and the super-twisting control law is formed. Then the total disturbance estimated by the observer is compensated in the control law and the final control signal enters the dynamic model of six degrees of freedom of the aircraft. The output of the aircraft dynamic model is then fed back to the observer to complete the control loop.

Using Eq. (13), and (20) control law is defined as follows:

$$u = \frac{1}{b_0}(u_0 - z_3) \quad (26)$$

All disturbances and uncertainties are estimated in the non-linear extended state observer through z_3 and compensated for online in the control law. This step transforms a problem that traditionally belongs to system identification to that of disturbance rejection.

Stability analysis of Closed-Loop dynamics

Convergence analysis of Non-linear ESO

To evaluate the closed-loop stability of the observer, it is essential to derive the error dynamics of the non-linear extended state observer. By analyzing Eqs. (11) and (13), the dynamic error equation for the non-linear ESO is formulated as follows:

$$\begin{cases} \dot{e}_1 = e_2 - l_1 e_1 \\ \dot{e}_2 = e_3 - l_2 e_1 \\ \dot{e}_3 = -l_3 f_{al}(e_1) - H(t) \end{cases} \quad (27)$$

By definition $\eta_i(t) = \frac{e_i(t)}{\omega_0^{i-1}}$, $i = 1, 2, 3$, Eq. (27) is rewritten as follows:

$$\begin{cases} \dot{\eta}_1 = \omega_0 \eta_2 - \frac{l_1}{\omega_0} \eta_1 \\ \dot{\eta}_2 = \omega_0 \eta_3 - \frac{l_2}{\omega_0} \eta_1 \\ \dot{\eta}_3 = \frac{H(t)}{\omega_0^2} - \frac{l_3}{\omega_0^2} f_{al}(\eta_1) \end{cases} \quad (28)$$

Equation (28) can be expressed as:

$$\dot{\eta} = \omega_0 A \eta + B \frac{H(t)}{\omega_0} \quad (29)$$

where .

$$A = \begin{bmatrix} -a_1 & 1 & 0 \\ -a_2 & 0 & 1 \\ -a_3 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Theorem1 If it is assumed that $H(t)$ is bounded. There exist a positive constant q_i , $i = 1, 2, 3$ and a finite time $T > 0$ such that $|e_i| \leq q_i$, $i = 1, 2, 3$, $\forall t \geq T$, $\omega_0 > 0$.

Proof Solving (29), the following equation can be obtained:

$$\eta(t) = e^{\omega_0 A t} \eta(0) + \int_0^t e^{\omega_0 A(t-\tau)} B \frac{H(\tau)}{\omega_0^2} d\tau \quad (30)$$

Let

$$q(t) = \int_0^t e^{\omega_0 A(t-\tau)} B \frac{H(\tau)}{\omega_0^2} d\tau, \eta(t) = e^{\omega_0 A t} \eta(0) + \int_0^t e^{\omega_0 A(t-\tau)} B \frac{H(\tau)}{\omega_0^2} d\tau \quad (31)$$

Since $|H(t)| \leq \xi$ and $(e^{\omega_0 A(t-\tau)} B)_i > 0$, $i = 1, 2, 3$ then

$$\begin{aligned} q_i(t) &= \int_0^t (e^{\omega_0 A(t-\tau)} B)_i \frac{H(\tau)}{\omega_0^2} d\tau \\ &= \frac{1}{\omega_0^2} \int_0^t (e^{\omega_0 A(t-\tau)} B)_i H(\tau) d\tau \\ &\leq \frac{\xi}{\omega_0^2} \int_0^t (e^{\omega_0 A(t-\tau)})_i d\tau \\ &= \frac{\xi}{\omega_0^2} (e^{\omega_0 A t})_i \int_0^t (e^{-\omega_0 A \tau})_i d\tau \\ &= \frac{\xi}{\omega_0^2} (e^{\omega_0 A t})_i \left(\frac{1}{-\omega_0 A} \right)_i (e^{-\omega_0 A \tau} B)_i \Big|_0^t \\ &= \frac{\xi}{\omega_0^2} [(-A^{-1} B)_i + (A^{-1} e^{\omega_0 A t} B)_i], i = 1, 2, 3 \end{aligned} \quad (32)$$

Then

$$|q_i(t)| \leq \frac{\xi}{\omega_0^2} [|(-A^{-1} B)_i| + |(A^{-1} e^{\omega_0 A t} B)_i|], i = 1, 2, 3 \quad (33)$$

And it can be calculated:

$$|(A^{-1}B)_i| = \begin{cases} 1, & i = 1, \\ 3, & i = 2, 3. \end{cases} \quad (34)$$

According to A is a Hurwitz matrix, there exist a finite time $T > 0$, that for all $t \geq T$,

$$\begin{cases} |(e^{\omega_0 A t})_{i,j}| \leq \frac{1}{\omega_0^3}, & i, j = 1, 2, 3, \\ |(e^{\omega_0 A t} B)_i| \leq \frac{1}{\omega_0^3}, & i = 1, 2, 3 \end{cases} \quad (35)$$

Equation (32) can be written as follows:

$$|\eta_i| \leq |[e^{\omega_0 A t} \eta(0)]_i| + |q_i| \quad (36)$$

Let $\varepsilon_s(0) = \varepsilon_1(0) + \frac{|\varepsilon_2(0)|}{\omega_0^2} + \frac{|\varepsilon_3(0)|}{\omega_0^3}$, and $\eta_i(t) = \frac{\varepsilon_i(t)}{\omega_0^{i-1}}$, then :

$$|\varepsilon_i(t)| \leq \left| \frac{\varepsilon_s(0)}{\omega_0^3} \right| + \frac{3R}{\omega_0^{4-i}} + \frac{4R}{\omega_0^{7-i}} = q_i \quad (37)$$

This result guarantees the finite-time convergence of the observer.

Controller stability analysis

To prove the stability of the controller using the Lyapunov method, a positive definite function $V > 0$ is considered. This function is continuous and differentiable, with $V(0) = 0$ at the equilibrium point. If it can be proven that $\dot{V} < 0$ for all $t > 0$ and all states except the equilibrium, then it can be concluded that the system is asymptotically stable. This implies that the system states will converge to the equilibrium point as time approaches infinity. The general super-twisting control law is considered as follows:

$$U = g(x)^{-1}(\ddot{x}_d - \lambda \dot{\varepsilon} - k_1 |s|^{\frac{1}{2}} \text{sign}(s) - k_2 \int_0^t \text{sign}(s(\tau)) d\tau - f(x)) \quad (38)$$

Assumption 1 All the state variables of the 6-degree-of-freedom of aircraft are either directly measurable or can be estimated through observation.

Assumption 2 The reference trajectories and their first and second derivatives are bounded.

Assumption 3 $f(x)$ is bounded, so that $|f(x)| \leq D$.

Proof To analyze the stability the following Lyapunov function is proposed:

$$V = \frac{1}{2} s^2 \quad (39)$$

Theorem 2 For the system to be stable, the time derivative of V must be negative definite ($\dot{V} \leq 0$):

$$\begin{aligned} \dot{V} &= s \dot{s} \\ &= s(\ddot{\varepsilon} + \lambda \dot{\varepsilon}) = s(g_u u + f(x) - \ddot{x}_d^1 + \lambda \dot{\varepsilon}) \end{aligned} \quad (40)$$

By incorporating the ST-SMC law and the estimates from the non-linear extended state observer, and according to $\frac{g_x}{b_0} \sim 1$, $z_3 = \hat{f}(x)$, and combining Eqs. (26), (38), (40):

$$\begin{aligned} \dot{V} &= s(\ddot{x}_d^1 - \lambda \dot{\varepsilon} - k_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) - k_2 \int_0^t \text{sign}(\hat{s}(\tau)) d\tau - \hat{f}(x) + f(x) - \ddot{x}_d^1 + \lambda \dot{\varepsilon}) \\ &= s(-k_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) - k_2 \int_0^t \text{sign}(\hat{s}(\tau)) d\tau + \tilde{f}(x) + \lambda \dot{\varepsilon}) \\ \dot{V} &= s(\ddot{x}_d^1 - \lambda \dot{\varepsilon} - k_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) - k_2 \int_0^t \text{sign}(\hat{s}(\tau)) d\tau - \hat{f}(x) + f(x) - \ddot{x}_d^1 + \lambda \dot{\varepsilon}) \\ &= s(-k_1 |\hat{s}|^{\frac{1}{2}} \text{sign}(\hat{s}) - k_2 \int_0^t \text{sign}(\hat{s}(\tau)) d\tau + \tilde{f}(x) + \lambda \dot{\varepsilon}) \\ &= s(-k_1 |\lambda \tilde{\varepsilon} - \dot{\tilde{\varepsilon}} + s|^{\frac{1}{2}} \text{sign}(-\lambda \tilde{\varepsilon} - \dot{\tilde{\varepsilon}} + s) - k_2 \int_0^t \text{sign}(-\lambda \tilde{\varepsilon} - \dot{\tilde{\varepsilon}} + s(\tau)) d\tau + \tilde{f}(x) + \lambda \dot{\tilde{\varepsilon}}) \end{aligned} \quad (41)$$

Where $\tilde{\varepsilon}, \dot{\tilde{\varepsilon}}$ are observer errors and their derivative. Referring to the results established in Theorem 1, the NESO ensures that the observer errors remain bounded over time. Consequently, Eq. (40) can be expressed as follows:

$$\dot{V} \approx s(-k_1 (|s|^{\frac{1}{2}} \text{sign}(s) - k_2 \int_0^t \text{sign}(s(\tau)) d\tau) \quad (42)$$

Parameters	Value	Parameters	Value
W	13000 (lbs)	$(C_{L_0}, C_{L_u}, C_{L_\alpha}, C_{L_{\dot{\alpha}}}, C_{L_q})$	(1.2, 0.04, 5.04, 1.6, 4.1)
(I_{xx}, I_{yy}, I_{zz})	(28000, 18800, 47000) (slugft ²)	$(C_{m_0}, C_{m_u}, C_{m_\alpha}, C_{m_q})$	(0.047, −0.01, −0.66, −5, 13.5)
I_{xz}	1300 (slugft ²)	$(C_{l_\beta}, C_{l_p}, C_{l_r})$	(−0.173, −0.390, 0.450)
S	230 (ft ²)	$(C_{y_\beta}, C_{y_p}, C_{y_r})$	(−0.730, 0, 0.4)
c	7 (ft)	$(C_{n_\beta}, C_{n_p}, C_{n_r})$	(0.15, −0.130, −0.260)
b	34 (ft)	$(C_{D_{\delta_E}}, C_{L_{\delta_E}}, C_{M_{\delta_E}})$	(0, 0.4, −0.98)
U_1	170 (ft/sec)	$(C_{l_{\delta_A}}, C_{l_{\delta_R}})$	(0.149, 0.014)
$\bar{q}$	0.0023768924 (lbs/ft ²)	$(C_{y_{\delta_A}}, C_{y_{\delta_R}})$	(0, 0.140)
$(C_{D_0}, C_{D_u}, C_{D_\alpha})$	(1.2, 0.04, 5.04, 1.6, 4.1)	$(C_{n_{\gamma_A}}, C_{n_{\delta_R}})$	(−0.05, −0.074)

Table 2. Aircraft specification.

Parameters	Value
(u_0, v_0, w_0)	(170, 0, 14) (ft/sec)
(p_0, q_0, r_0)	(0, 0, 0) (deg/s)
$(\phi_0, \theta_0, \psi_0)$	(0, 3, 0) (deg)
(x_0, y_0, z_0)	(0, 0, −1000) (ft)

Table 3. Initial conditions.

If $k_1^2 > 4k_2$, $k_2 > LI_3$, then $\dot{V} < 0$, therefore, according to Lyapunov’s stability theorem, the closed-loop system is asymptotically stable.

Simulation result

The aircraft’s parameter values are presented in Table 2²².
The initial conditions for simulation are presented in Table 3.

Disturbance modeling

In the design and analysis of aircraft control systems, it is essential to account for various sources of disturbances and uncertainties that may degrade performance and compromise stability. This section presents a comprehensive model of disturbances, encompassing multiple sources of errors and uncertainties that can impact the aircraft’s flight dynamics and control system performance.

1) **Actuator Dynamics:** The first source of error considered is the imperfect performance of actuators. Actuator dynamics can introduce disturbances in the control inputs, potentially leading to deviations from the desired control signals. This disturbance is modeled as an additive term to the control inputs, representing the discrepancy between the commanded and actual actuator outputs.

2) **Model Uncertainties:** The second source of error stems from the simplifying assumptions made in deriving the equations of motion. Several flight parameters, such as mass properties, moments of inertia, and thrust forces, are typically assumed constant for simplification. These uncertainties are modeled as percentage variations in the respective parameters.

These uncertainties are modeled as percentage changes in the respective parameters, allowing for both increases and decreases in their values.

3) **Sensors Bias:** Sensor outputs, which serve as inputs to the controller and observer, can be subject to noise and errors. This source of disturbance is modeled as sensor bias with a constant noise of 2° . The bias is applied to the feedback of Euler angles and their rates, representing systematic errors in the sensor measurements.

4) **Atmospheric Disturbances:** During flight, an aircraft is subjected to wind and atmospheric turbulence, which can significantly disrupt the attitude control system. To accurately model these atmospheric disturbances, the Dryden wind model is utilized. This model is derived from atmospheric turbulence velocity spectra and their gradients, forming a transfer function that converts white noise signals into colored noise. This conversion effectively simulates realistic wind patterns. The parameters of the Dryden model are tuned according to the MIL-F-8785 C standard, ensuring that the simulated atmospheric disturbances closely match real-world conditions. As illustrated in Fig. 7, the implemented Dryden wind model is designed based on the aircraft’s flight altitude, assuming light turbulence conditions. The resulting wind velocities are added to the aircraft’s velocities in all three directions.

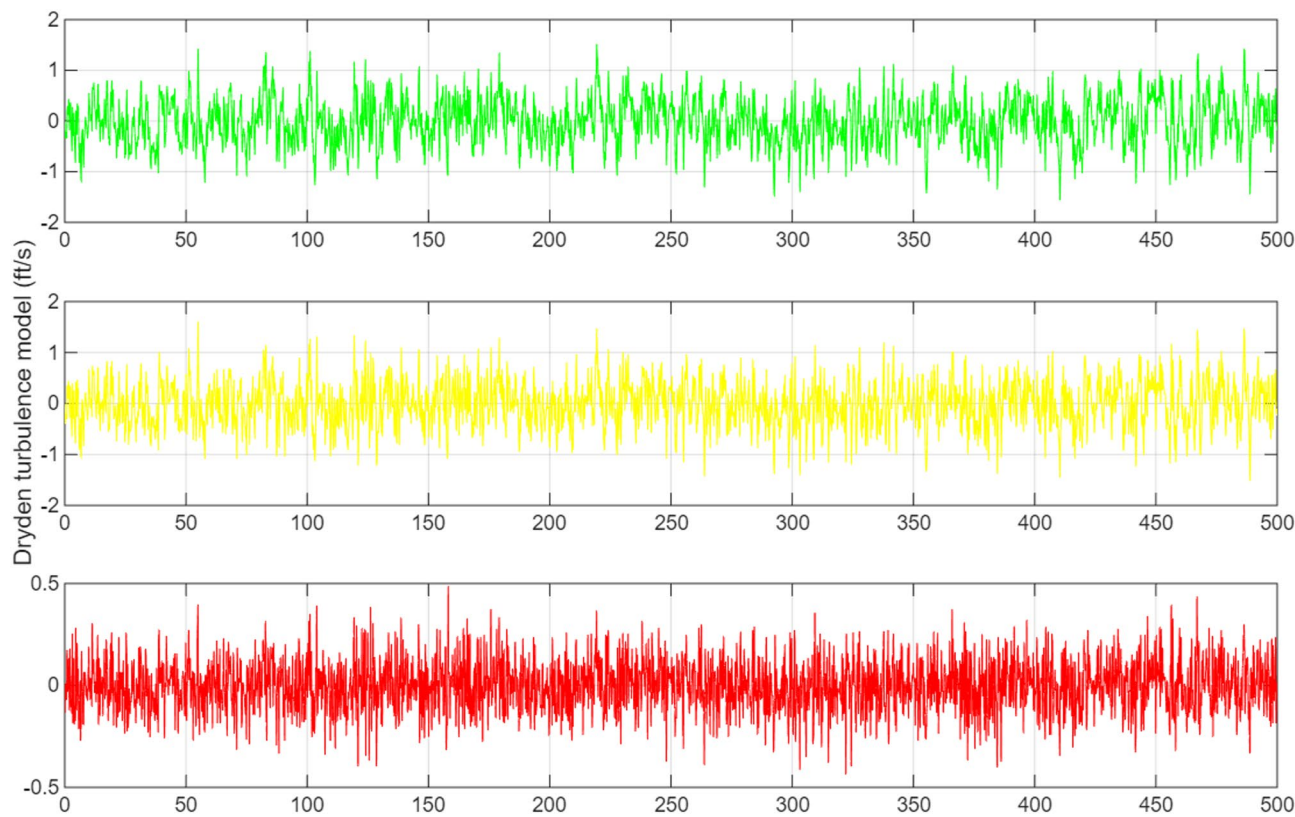


Fig. 7. Dryden turbulence wind model.

Source of Disturbance	Parameter	Value
Actuators	Amplitude	$0.01 * sign [\delta_a \quad \delta_e \quad \delta_r]^T$
Aerodynamic uncertainty	c_i	$\pm 20\%$
Mass and inertial uncertainty	$m, I_{xx}, I_{yy}, I_{zz}$	$\pm 20\%$
Sensors bias	Amplitude	2 deg
Atmospheric disturbances	Amplitude	$Light \text{ Turbulence, } \sigma_{g_{Light}} = 2.95 \text{ ft/sec}$

Table 4. Disturbance Sources.

By incorporating these four major sources of disturbances it is possible to achieve a comprehensive and realistic evaluation of the aircraft’s performance under various operating conditions. The modeled disturbances are summarized in Table 4.

Simulation results for the descent phase are presented in Figs. 8, 9, 10, 11, 12, 13, 14 and 15. Figures 8 and 9 show the Euler angles and the Euler rate angle. As illustrated in the response plots, the proposed method effectively eliminates disturbances and drives the system to the desired states. Unlike simple linear control methods, which often exhibit significant limitations in handling complex trajectories, the proposed approach demonstrates superior precision and robustness. A critical factor in achieving this performance is the adaptive tuning of the control gains through reinforcement learning. Instead of being manually adjusted, the gains are automatically optimized during interaction with the environment, allowing the controller to balance convergence speed, robustness, and control effort in real-time. According to Figs. 8 and 9, the reinforcement learning agent successfully adjusts λ to accelerate convergence without excessive overshoot, while k_1 and k_2 are regulated to enhance tracking performance and mitigate chattering. This adaptive strategy ensures that the system output rapidly converges to the desired trajectory. Also, as illustrated in Fig. 12, the control effort is kept within allowable bounds. Considering that the model has six degrees of freedom, with attitude control implemented over three degrees of freedom, the stability of the system states is of paramount importance. Figure 10 presents the linear velocities, and Fig. 11 shows the position profiles during the descent phase, confirming the stability of the six-degree-of-freedom model under attitude control. Figure 13 shows the stability of the system’s internal dynamics. These graphs illustrate how the total velocity, angle of attack, and sideslip angle change throughout the maneuver.

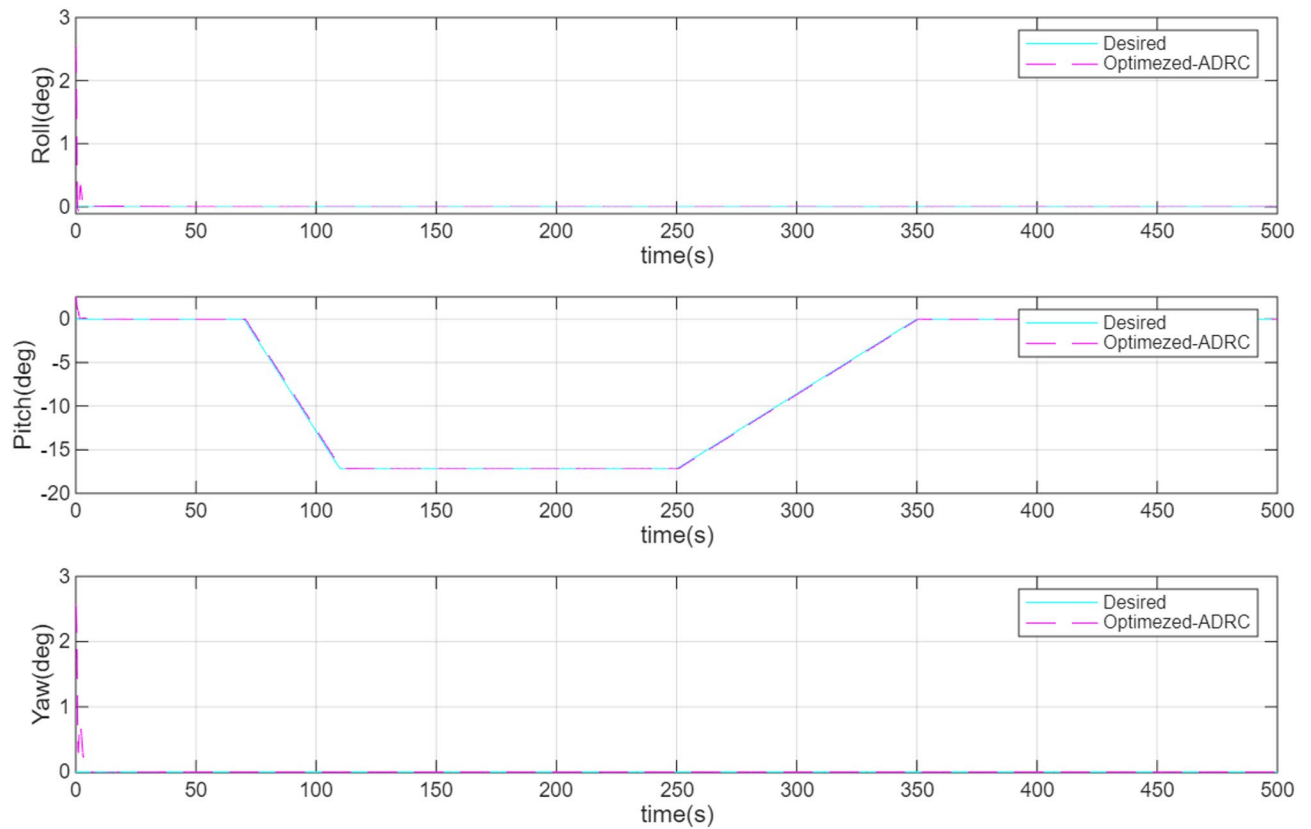


Fig. 8. Euler angles.

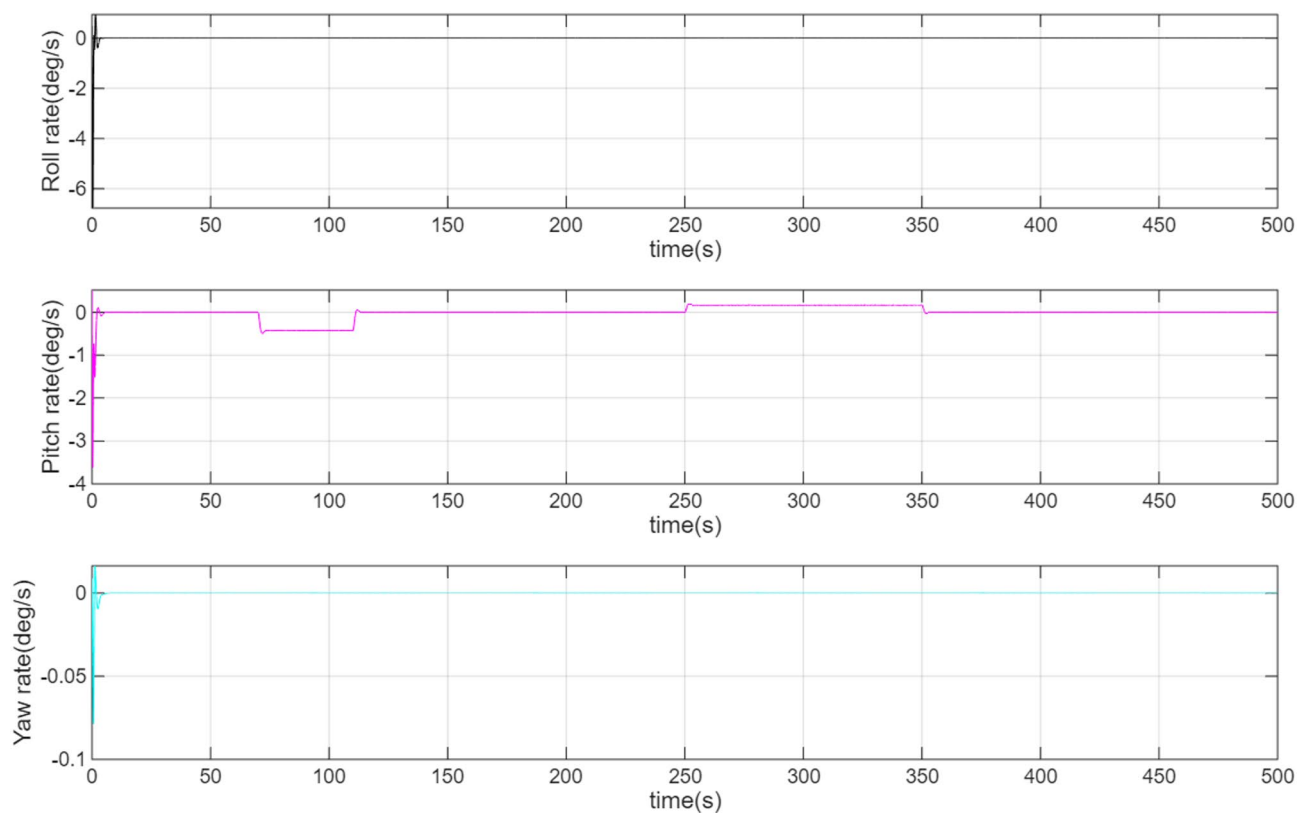


Fig. 9. Euler rate angles.

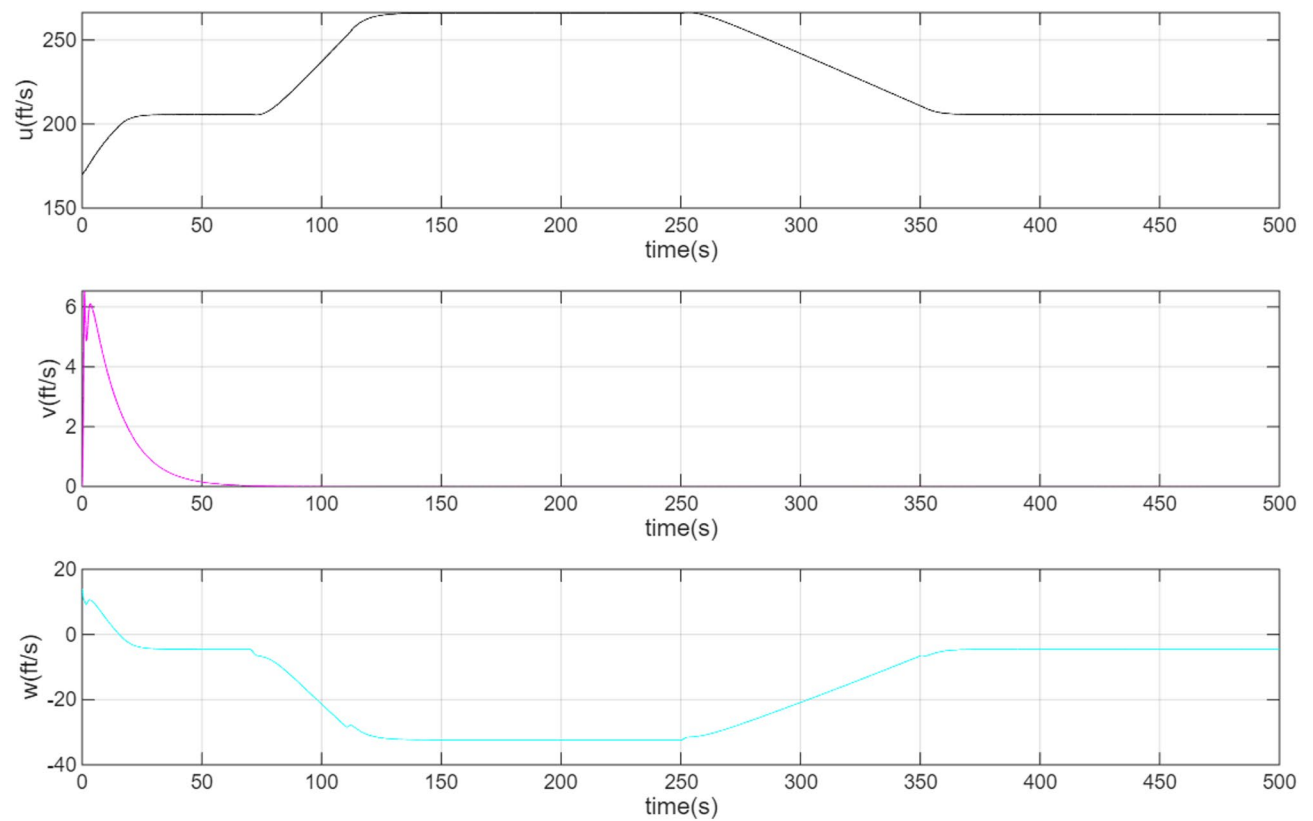


Fig. 10. Linear velocities.

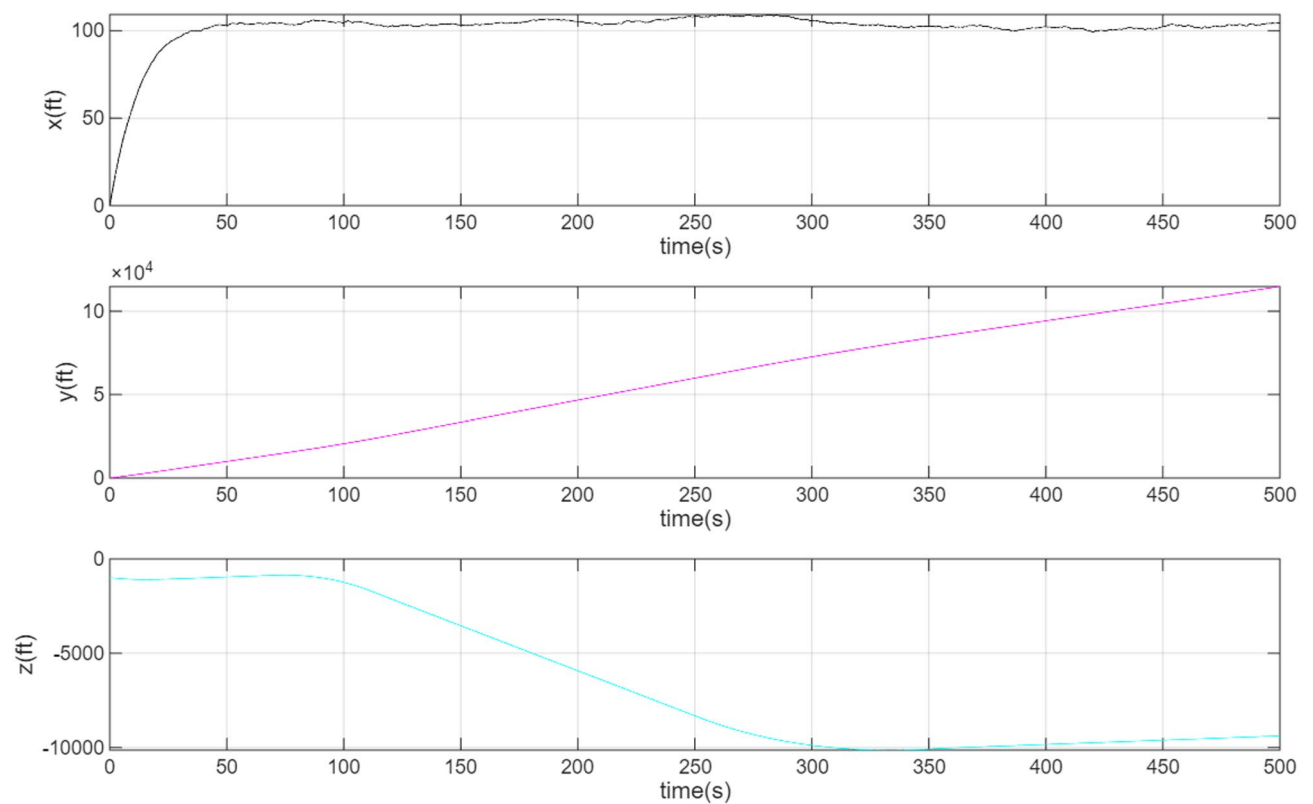


Fig. 11. Aircraft position.

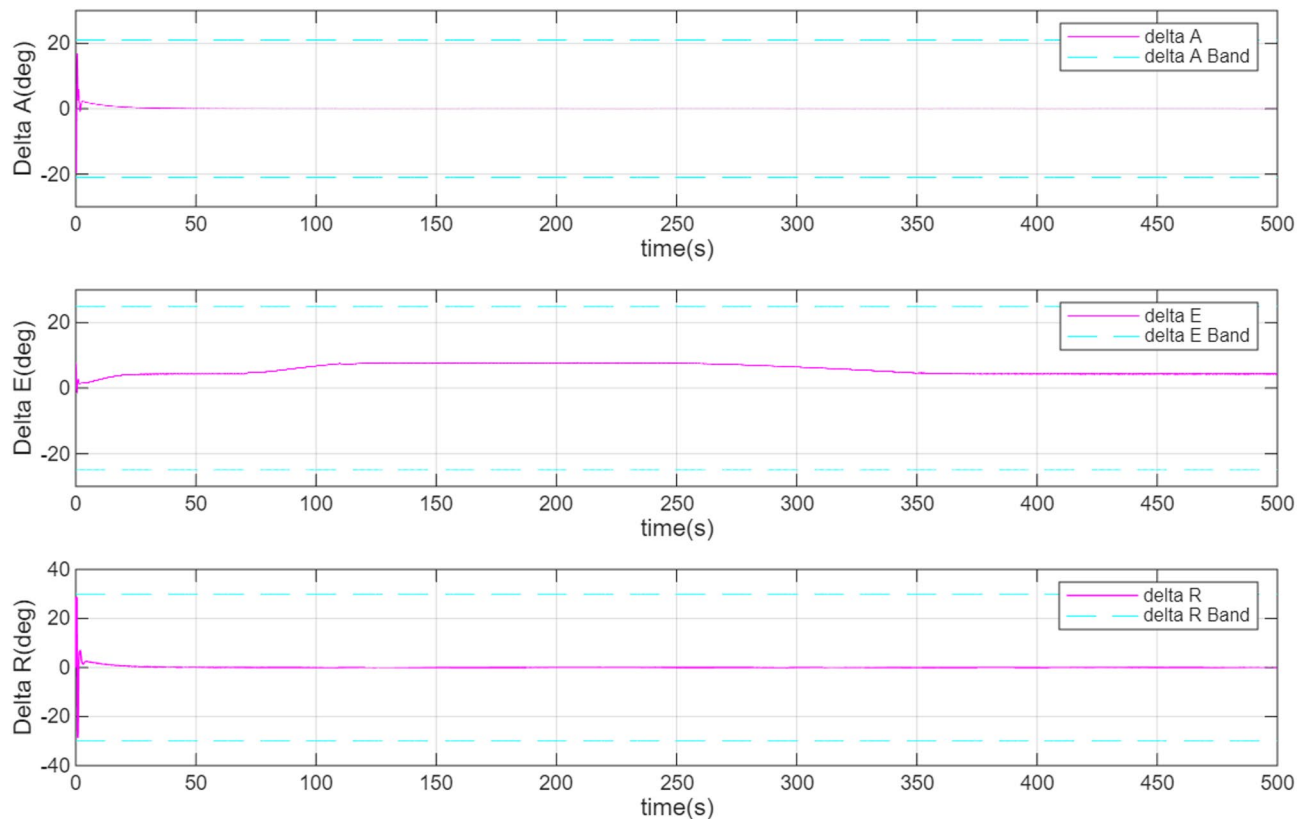


Fig. 12. Control efforts.

The gains of the NESO significantly impact the estimation of system states, uncertainties, and disturbances. The accuracy and effectiveness of the designed observer are validated by Figs. 14 and 15, which depict a precise estimation of Euler angles, their rates, and the total disturbance. This observer not only estimated the system state variables effectively but also captured external disturbances and uncertainty.

The total maneuver time, considering the path designed for the pitch angle, is set at 500 s, demonstrating the long-term stability and robustness of the controller and the observer.

The results demonstrate the effectiveness of the optimized-ADRC-based attitude control in stabilizing the six-degree-of-freedom model.

Evaluation and comparison of control methods

Two comparative analyses are conducted to demonstrate the proposed method's superiority. Firstly, the controller performance is evaluated against two reference methods: a standalone super-twisting sliding mode controller and a linear ADRC scheme, where the super-twisting controller is combined with a linear extended state observer (LESO). The latter is referred to as linear ADRC due to the linear nature of its observer. Then, the estimation capabilities of the nonlinear extended state observer are benchmarked against those of a linear extended state observer, providing a clear demonstration of its enhanced performance. These comprehensive comparisons clearly demonstrate the effectiveness and advantages of the proposed method, establishing it as a robust and efficient solution for attitude control in nonlinear systems. The simulation results are illustrated in Figs. 16, 17, 18 and 19.

Figure 16 illustrate the attitude control trajectories, comparing three control methods. The Optimized-ADRC method demonstrates the fastest convergence, bringing the system to the desired state in the shortest time. In contrast, the LADRC method shows slightly slower convergence, while the STA exhibits the slowest response among the three methods. In some cases, such as the overshoot and undershoot value, the nonlinear active disturbance rejection method has a better performance than the other two methods. The super-twisting controller, without an observer, has the simplest control loop structure. In contrast, Optimized-ADRC and LADRC rely on their respective observers for angle feedback, which inherently increases the complexity of the controller components. Figures 17, 18 and 19 depict the control effort for each method. As shown in Fig. 20, the STA method shows significant jumps in the elevator control input at peak points of the tracking path. This is attributed to the lack of a tracking differentiator from which the differential signal is extracted, potentially degrading the control method's performance. But in the other two methods, due to the presence of tracking accuracy, there are no such jumps. Due to the control law design based on observer feedback, the maximum control effort required for Optimized-ADRC and LADRC is higher than that of the STA method. This

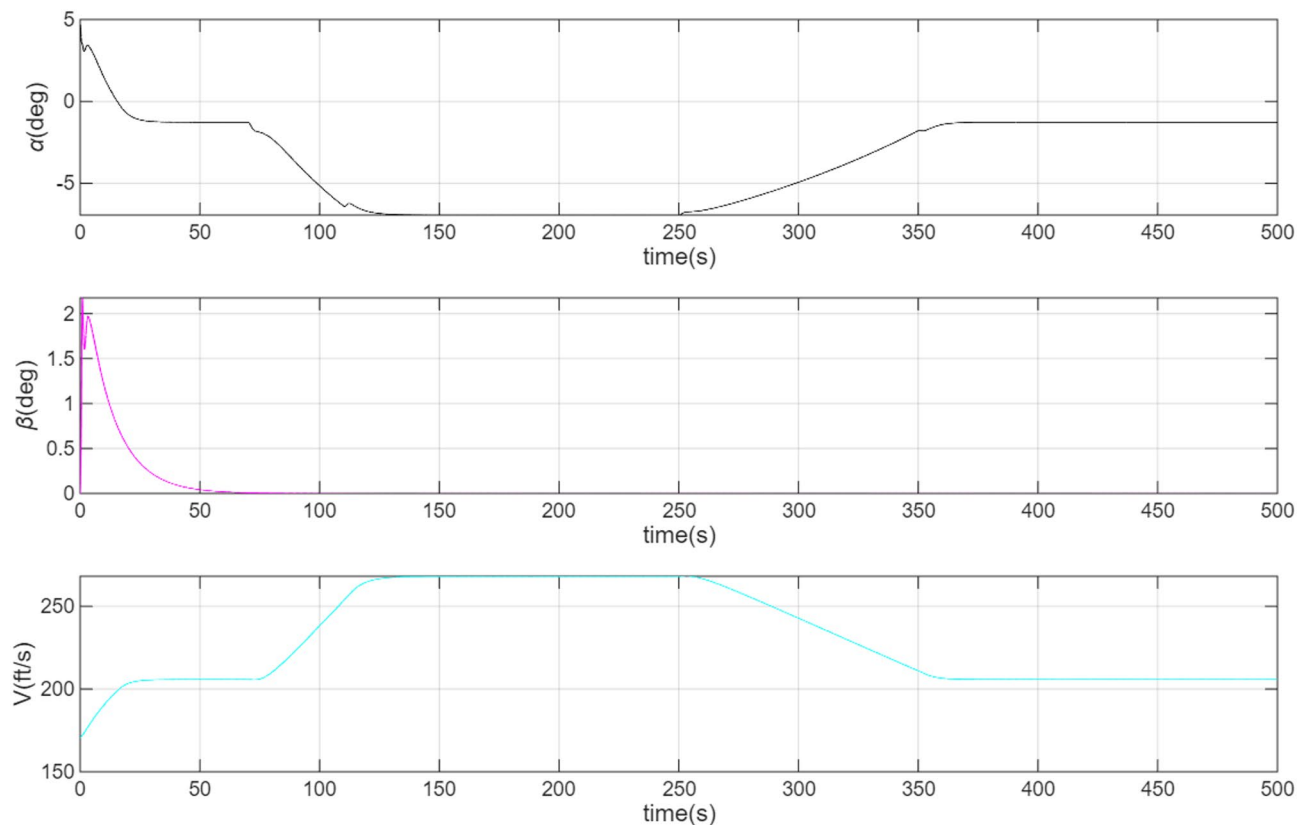


Fig. 13. α , β , V .

difference arises because the STA method derives control signals from the system without utilizing observer-based estimations.

The three control methods are systematically compared based on key performance metrics, including settling time, maximum control effort, mean squared error (MSE), integral absolute error (IAE), integral squared error (ISE), both in the presence and absence of external disturbances. These comparative results are presented in Tables 5 and 6, offering a comprehensive and detailed analysis of each method's performance, thereby providing valuable insights into their relative advantages and limitations.

Based on the results presented in Tables 5 and 6, the Optimezed-ADRC method demonstrates superior performance across multiple metrics. Specifically, Optimezed-ADRC exhibits the shortest settling time and the lowest maximum error among all evaluated methods, highlighting its robustness and efficiency in managing disturbances. In contrast, the STA shows the highest maximum error, indicating its relatively lower performance in disturbance rejection and trajectory tracking. The use of observers in ADRC methods offers several advantages, including effective mitigation of sensor noise, redundancy in the event of sensor failures, and cost optimization when employing high-precision sensors. Moreover, comparing the tables reveals that the error values for both ADRC methods remain relatively consistent regardless of the presence or absence of external disturbances. This stability is attributed to the accurate disturbance estimation and compensation achieved through the extended state observer. Overall, Optimezed-ADRC consistently demonstrates superior performance and less sensitivity to disturbances than other methods.

The next step of this study will compare the estimation accuracy of the NESO with the LESO, further assessing their respective capabilities in managing uncertainties and external disturbances.

Figures 20, 21, 22, 23, 24 and 25 present a comparative analysis of Euler angles and Euler rate angles estimation errors between NESO and LESO. The results demonstrate the superior performance of the NESO, characterized by reduced overshoot and faster stabilization, indicating more rapid and accurate estimation capabilities. This advantage is particularly evident in Fig. 19, where the NESO exhibits nearly zero estimation error at trajectory change points for the pitch angle, while the LESO shows noticeable errors. Furthermore, a comparison between Figs. 13 and 26 clearly illustrates the superior performance of the NESO in estimating total disturbances. Unlike the linear observer, the NESO provides significantly more accurate and closer approximations of the system dynamics and their associated disturbances, highlighting its robustness in handling complex and uncertain environments.

The comparative analysis of these two observers, conducted using error metrics both in the presence and absence of disturbances, is presented in Tables 7 and 8. This comprehensive evaluation further emphasizes the enhanced capabilities of the nonlinear ESO across a wide range of operational conditions, confirming its reliability and efficiency in control applications.

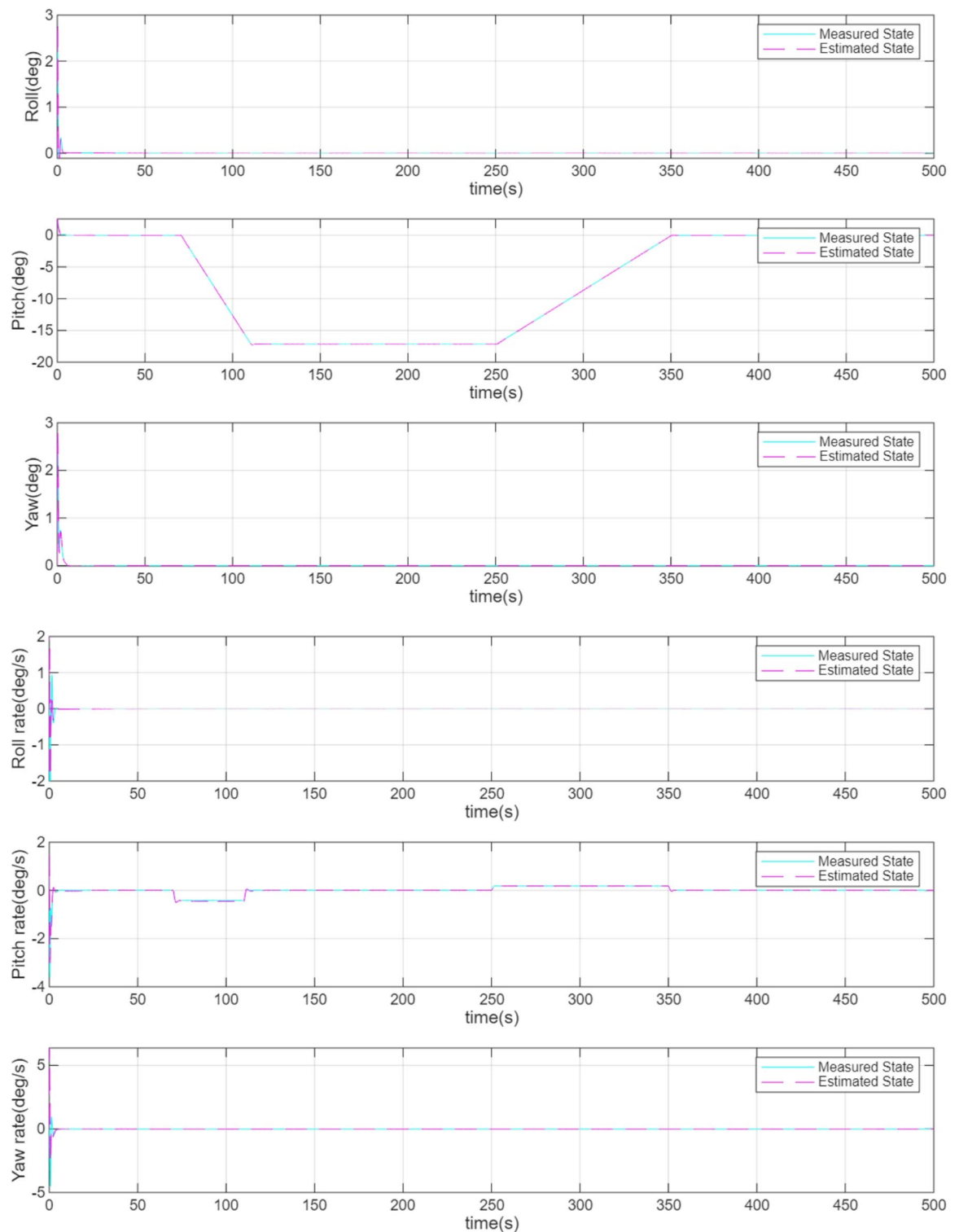


Fig. 14. Estimation of euler angles and euler rate angles.

Analyzing the error values presented in Tables 7 and 8, the NESO demonstrates superior performance in state estimation compared to the LESO. The NESO consistently exhibits lower overshoot and better overall performance across all evaluated scenarios. Moreover, it yields smaller values across different error metrics, further confirming its enhanced estimation capability.

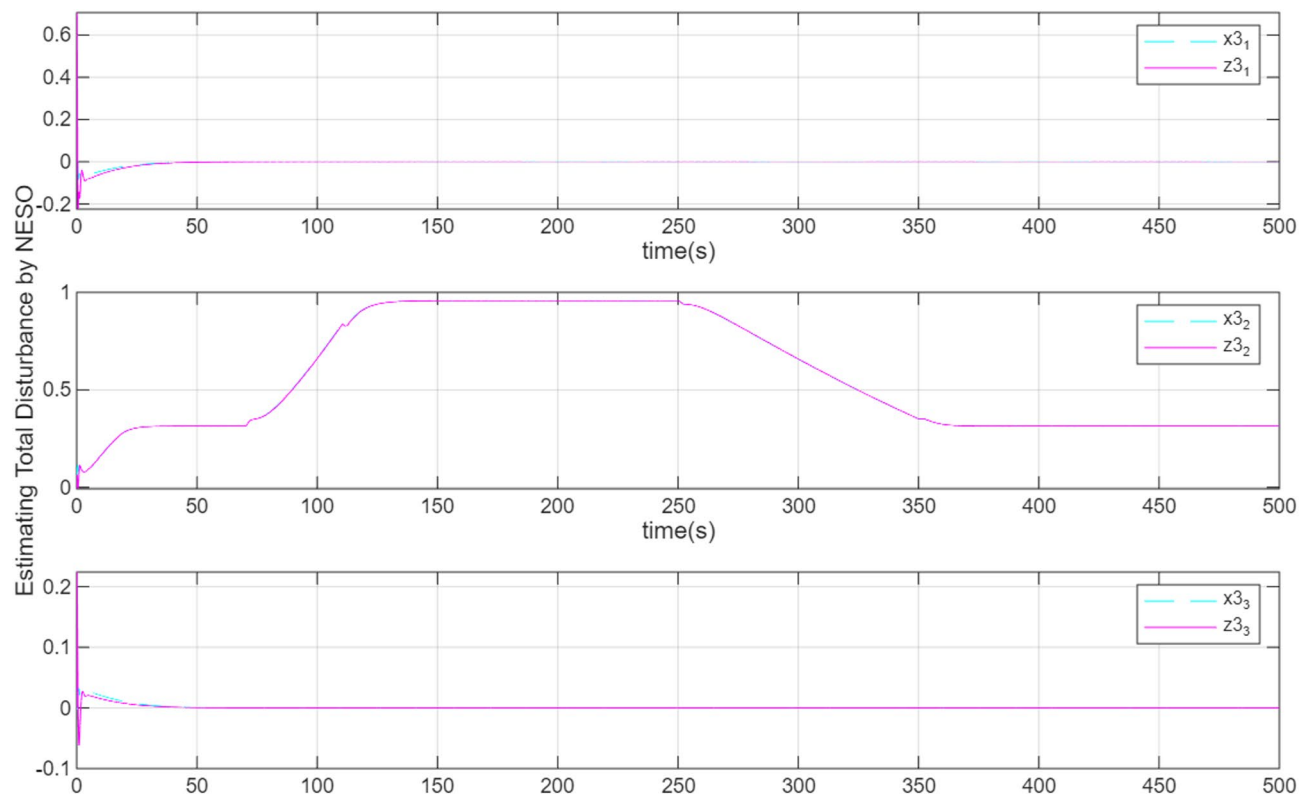


Fig. 15. Estimation total disturbance.

Conclusion

In this paper, a robust and efficient control strategy for attitude stabilization of a nonlinear aircraft model is presented by integrating active disturbance rejection control (ADRC) and an Optimezd super-twisting sliding mode control (ST-SMC) algorithm. The proposed control method effectively combines the disturbance estimation capability of the nonlinear extended state observer (NESO) with the robustness of the super-twisting algorithm and reinforcement learning to achieve precise attitude control in the presence of significant uncertainties and external disturbances. The stability analysis, performed using Lyapunov theory, confirms the asymptotic stability of the closed-loop system. Extensive simulation results, conducted on a Learjet 24 model considering the actuators model under various disturbance scenarios, validate the superior performance of the proposed method in terms of convergence speed, robustness, and disturbance rejection capability when compared with the linear ADRC scheme (based on STC+LESO) and the classical ST-SMC technique. The results demonstrate the proposed controller's capability to maintain stability even under severe disturbances and actuator faults, while achieving low tracking errors and high disturbance rejection efficiency. Furthermore, the comparative analysis between nonlinear and linear extended state observers (NESO and LESO) highlights the superior accuracy and faster convergence of the NESO. This research significantly contributes to modern flight control systems by providing a reliable and resilient approach to attitude stabilization, which can be effectively applied to both military and civilian aerospace applications.

Future research could focus on leveraging data-driven approaches for observer design, such as employing deep learning, which could significantly improve disturbance estimation accuracy and adaptive dynamics modeling. These methods can complement the NESO by learning complex and time-varying dynamics from data, thus enhancing the robustness of the control system.

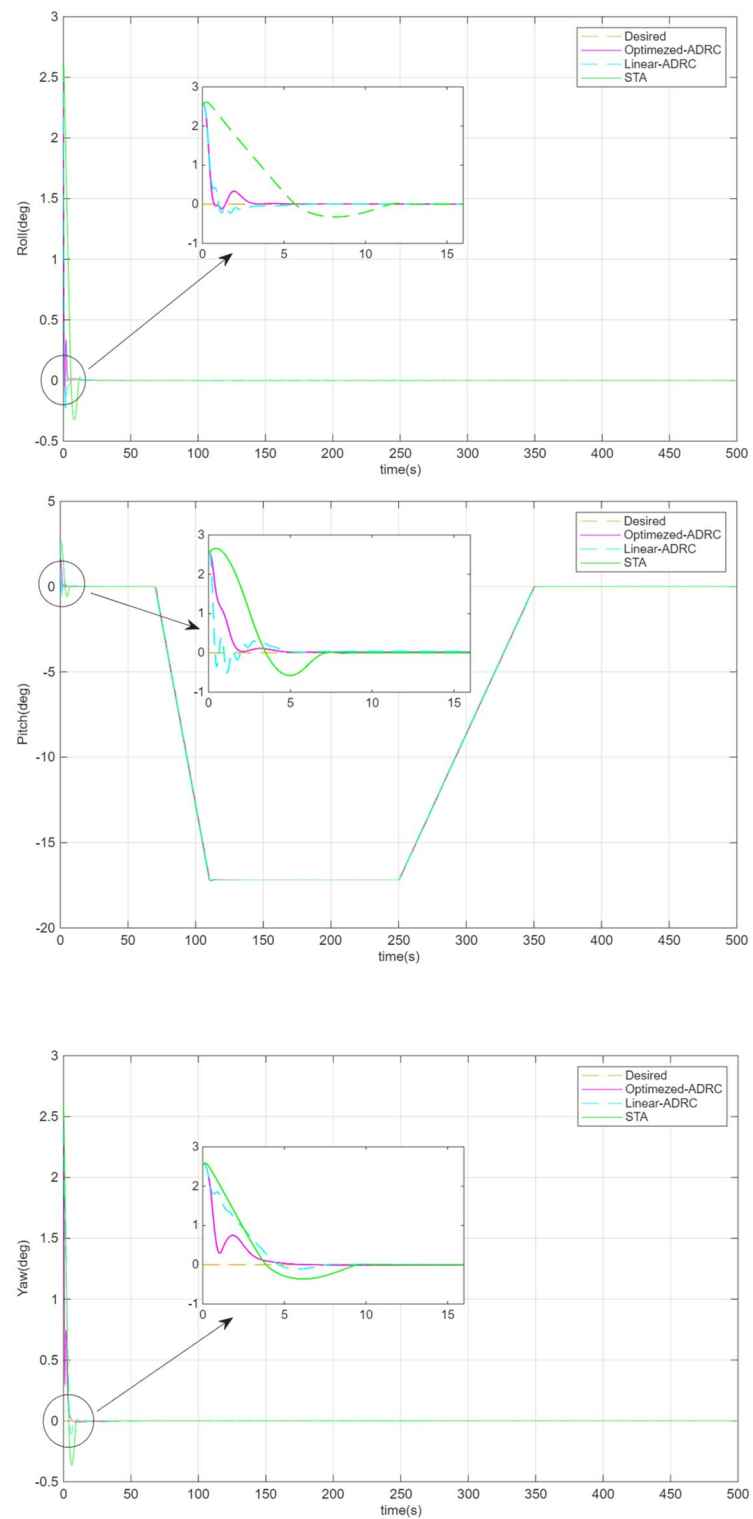


Fig. 16. Euler angles comparison.

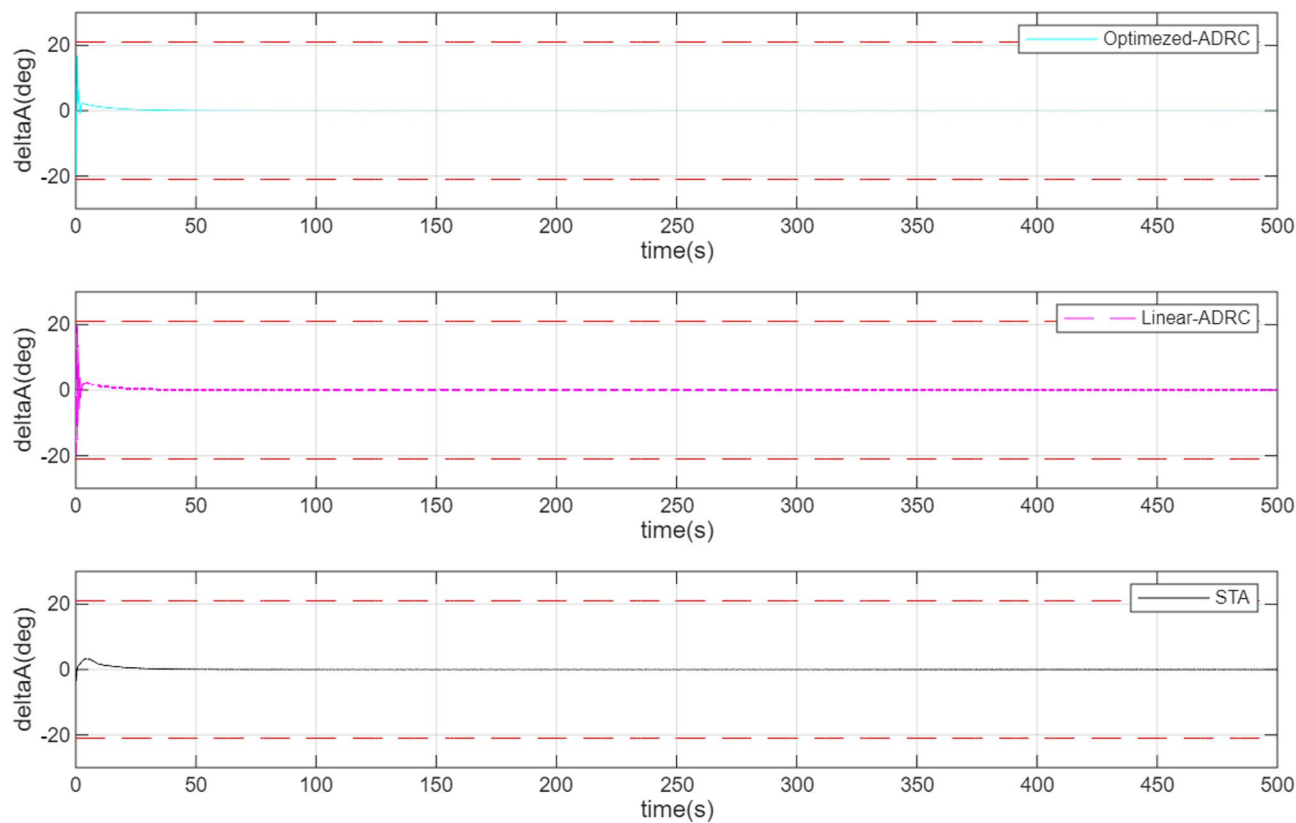


Fig. 17. Aileron control effort.

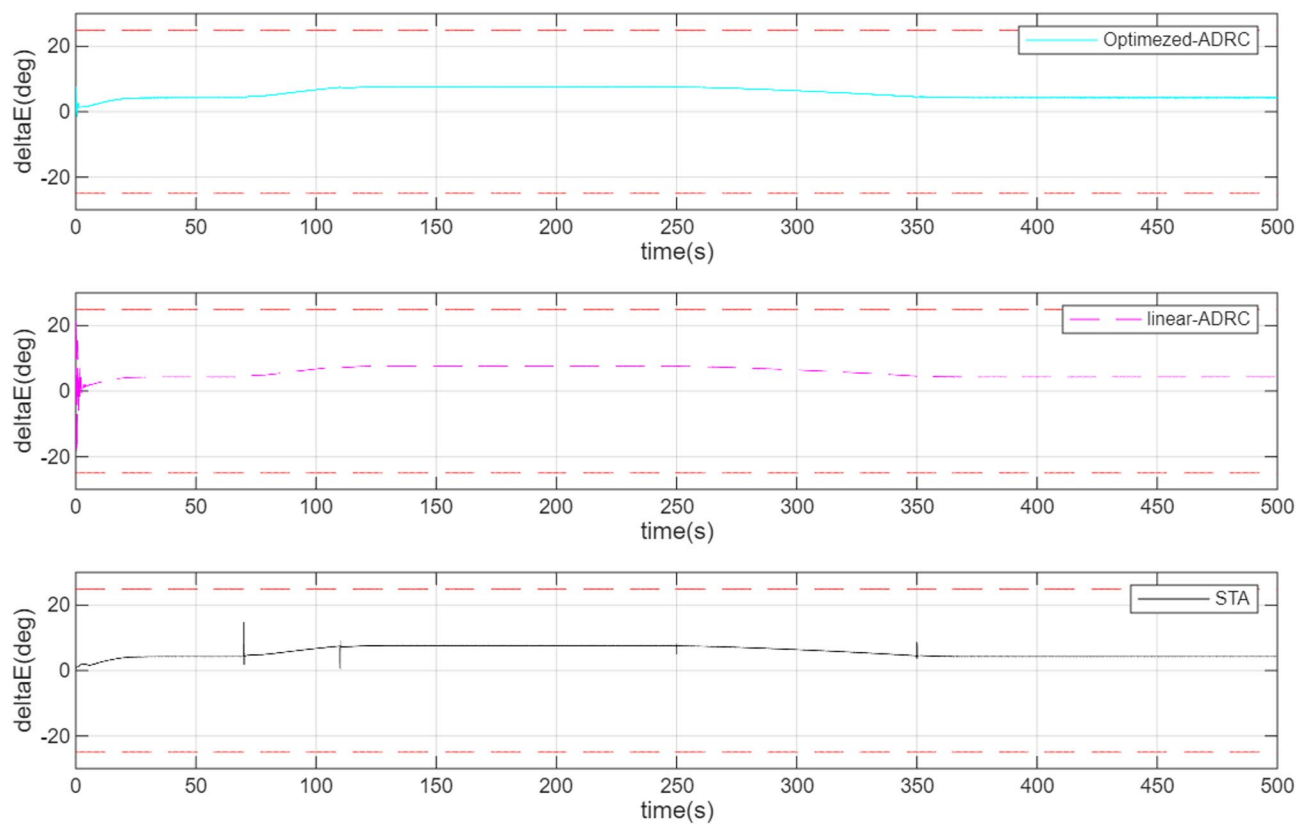


Fig. 18. Elevator control effort.

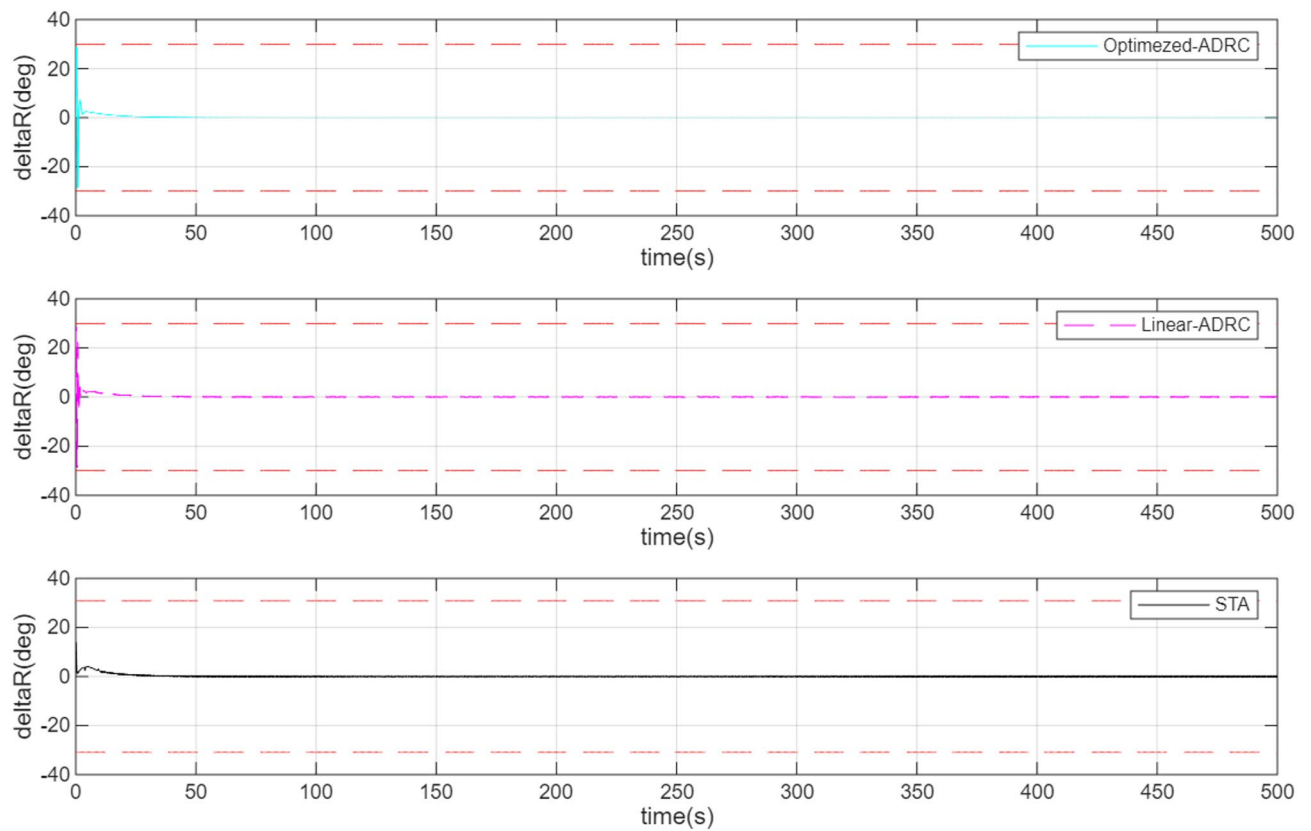


Fig. 19. Rudder control effort.

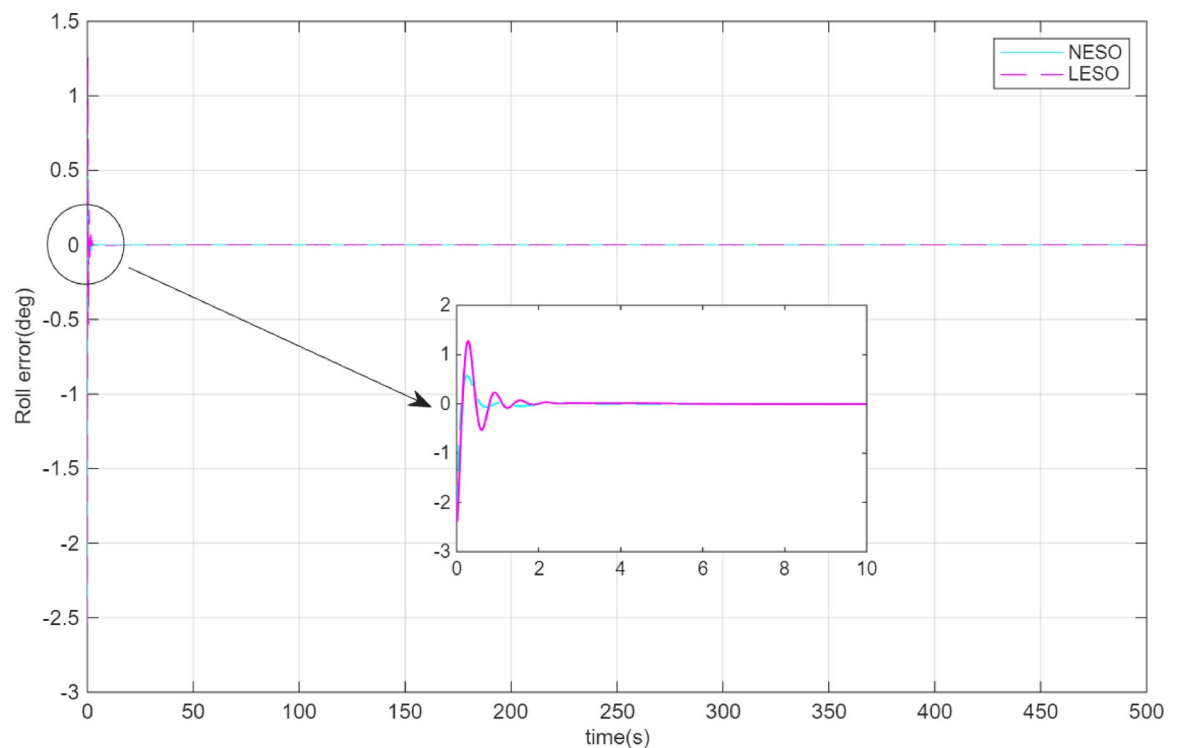


Fig. 20. Roll angle estimation error.

Controller	Optimized-ADRC	LADRC	ST-SMC
Settling Time (sec)	$T_{s\phi} = 2.25$ $T_{s\theta} = 5.24$ $T_{s\psi} = 4.54$	$T_{s\phi} = 5.52$ $T_{s\theta} = 7.02$ $T_{s\psi} = 7.13$	$T_{s\phi} = 10$ $T_{s\theta} = 6.25$ $T_{s\psi} = 8.13$
Maximum Control Effort (degree)	$\delta_A = 19.481$ $\delta_E = 7.7145$ $\delta_R = 28.684$	$\delta_A = 20.352$ $\delta_E = 20.926$ $\delta_R = 29.023$	$\delta_A = 2.6641$ $\delta_E = 17.248$ $\delta_R = 7.0384$
Mean Squared Error	$mse_\phi = 7.781e-07$ $mse_\theta = 3.2201e-06$ $mse_\psi = 1.0848e-06$	$mse_\phi = 8.5777e-07$ $mse_\theta = 5.0678e-06$ $mse_\psi = 1.6176e-06$	$mse_\phi = 4.4835e-06$ $mse_\theta = 8.6887e-06$ $mse_\psi = 3.2002e-06$
Integral Absolute Error	$Iae_\phi = 0.022361$ $Iae_\theta = 0.36416$ $Iae_\psi = 0.032125$	$Iae_\phi = 0.038454$ $Iae_\theta = 0.5379$ $Iae_\psi = 0.057465$	$Iae_\phi = 0.1055$ $Iae_\theta = 0.7214$ $Iae_\psi = 0.080204$
Integral Squared Error	$Ise_\phi = 0.00038272$ $Ise_\theta = 0.0016037$ $Ise_\psi = 0.0005361$	$Ise_\phi = 0.00062256$ $Ise_\theta = 0.0040277$ $Ise_\psi = 0.00080247$	$Ise_\phi = 0.0022354$ $Ise_\theta = 0.001838$ $Ise_\psi = 0.0015938$
Maximum Error	$max_{e_\phi} = 0.0356$ $max_{e_\theta} = 0.035604$ $max_{e_\psi} = 0.0356$	$max_{e_\phi} = 0.0356$ $max_{e_\theta} = 0.035622$ $max_{e_\psi} = 0.0366$	$max_{e_\phi} = 0.0456$ $max_{e_\theta} = 0.05427$ $max_{e_\psi} = 0.0491$

Table 5. Comparison of control methods under nominal conditions and in the presence of 20% uncertainty.

Controller	Optimized-ADRC	LADRC	ST-SMC
Settling Time (sec)	$T_{s\phi} = 2.96$ $T_{s\theta} = 4.54$ $T_{s\psi} = 5.66$	$T_{s\phi} = 5.43$ $T_{s\theta} = 5.57$ $T_{s\psi} = 8.34$	$T_{s\phi} = 11.53$ $T_{s\theta} = 7.02$ $T_{s\psi} = 9.06$
Maximum Control Effort (degree)	$\delta_A = 19.481$ $\delta_E = 7.6765$ $\delta_R = 28.648$	$\delta_A = 20.352$ $\delta_E = 22.918$ $\delta_R = 29.023$	$\delta_A = 3.4031$ $\delta_E = 14.74$ $\delta_R = 13.798$
Mean Squared Error	$mse_\phi = 1.33e-06$ $mse_\theta = 3.8622e-06$ $mse_\psi = 2.432e-06$	$mse_\phi = 1.3816e-06$ $mse_\theta = 4.8471e-06$ $mse_\psi = 4.6758e-06$	$mse_\phi = 8.7903e-06$ $mse_\theta = 8.0792e-06$ $mse_\psi = 6.0906e-06$
Integral Absolute Error	$Iae_\phi = 0.028205$ $Iae_\theta = 0.3746$ $Iae_\psi = 0.054151$	$Iae_\phi = 0.032562$ $Iae_\theta = 0.42037$ $Iae_\psi = 0.098119$	$Iae_\phi = 0.15844$ $Iae_\theta = 0.12653$ $Iae_\psi = 0.11728$
Integral Squared Error	$Ise_\phi = 0.00065519$ $Ise_\theta = 0.0019213$ $Ise_\psi = 0.0012062$	$Ise_\phi = 0.00098101$ $Ise_\theta = 0.0032878$ $Ise_\psi = 0.0023281$	$Ise_\phi = 0.0043854$ $Ise_\theta = 0.0070299$ $Ise_\psi = 0.0050355$
Maximum Error	$max_{e_\phi} = 0.044699$ $max_{e_\theta} = 0.044632$ $max_{e_\psi} = 0.044762$	$max_{e_\phi} = 0.045789$ $max_{e_\theta} = 0.054734$ $max_{e_\psi} = 0.054901$	$max_{e_\phi} = 0.055583$ $max_{e_\theta} = 0.076407$ $max_{e_\psi} = 0.065156$

Table 6. Comparison of control methods in the presence of external disturbances.

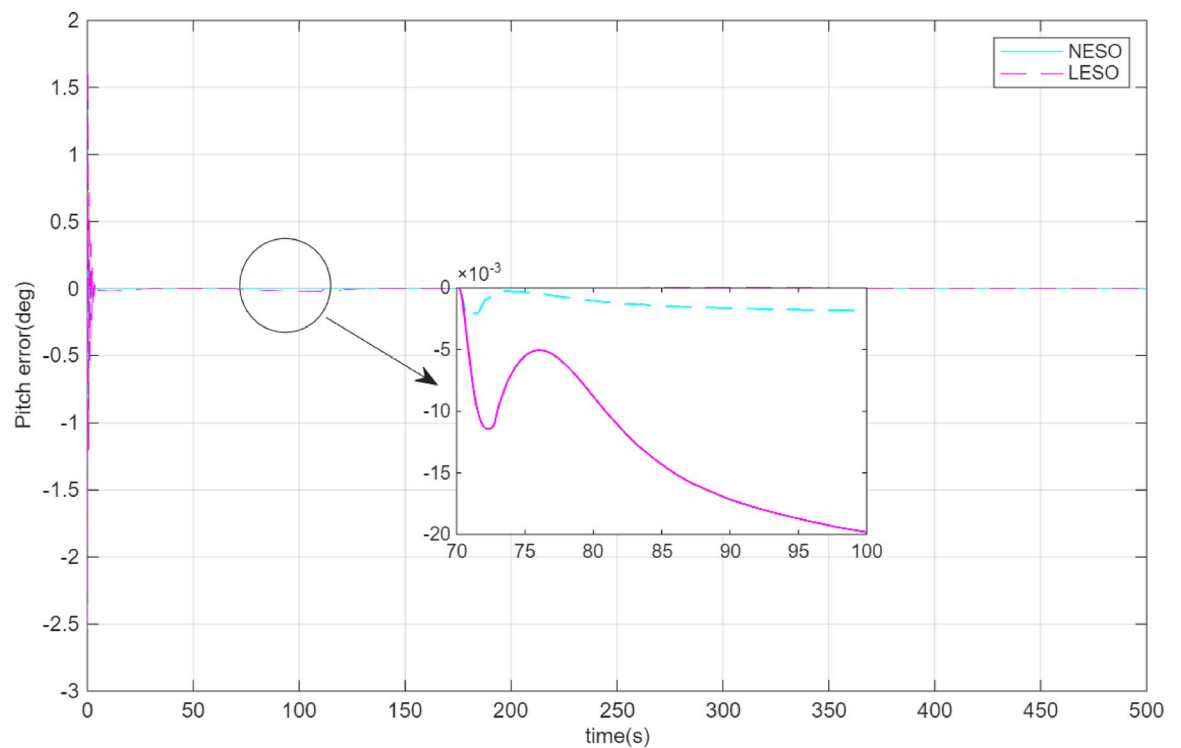


Fig. 21. Pitch angle estimation error.

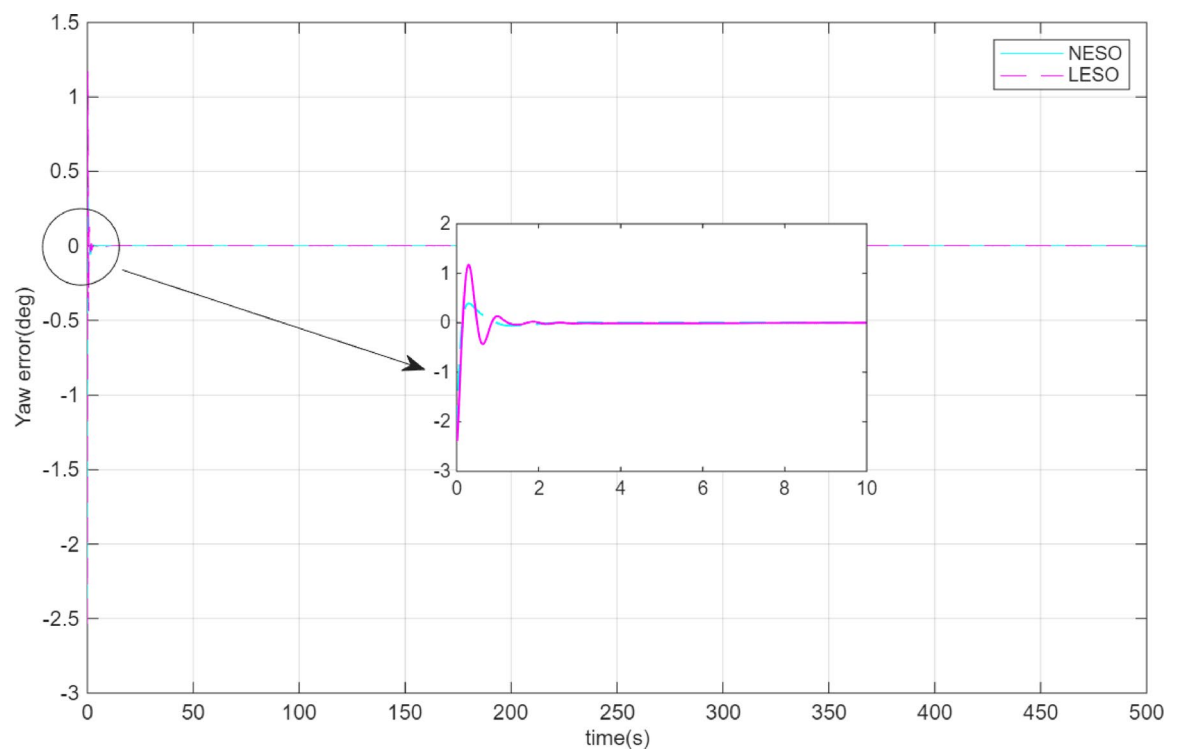


Fig. 22. Yaw angle estimation error.

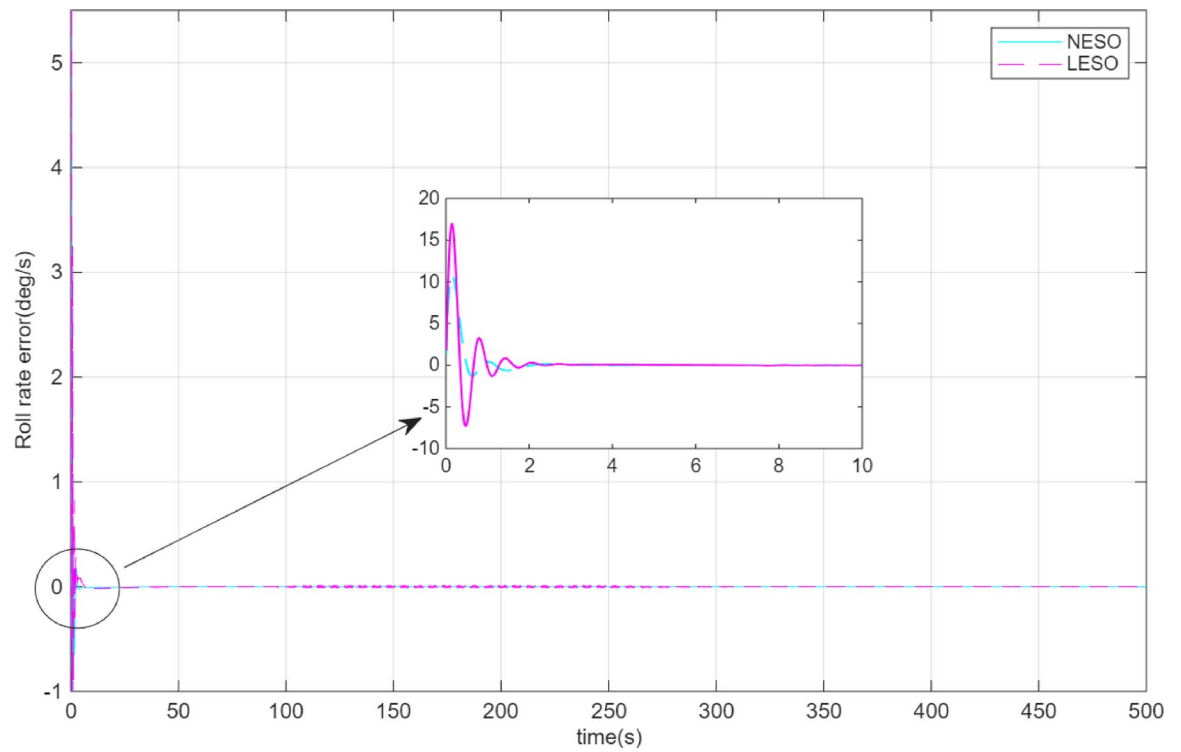


Fig. 23. Roll rate angle estimation error.

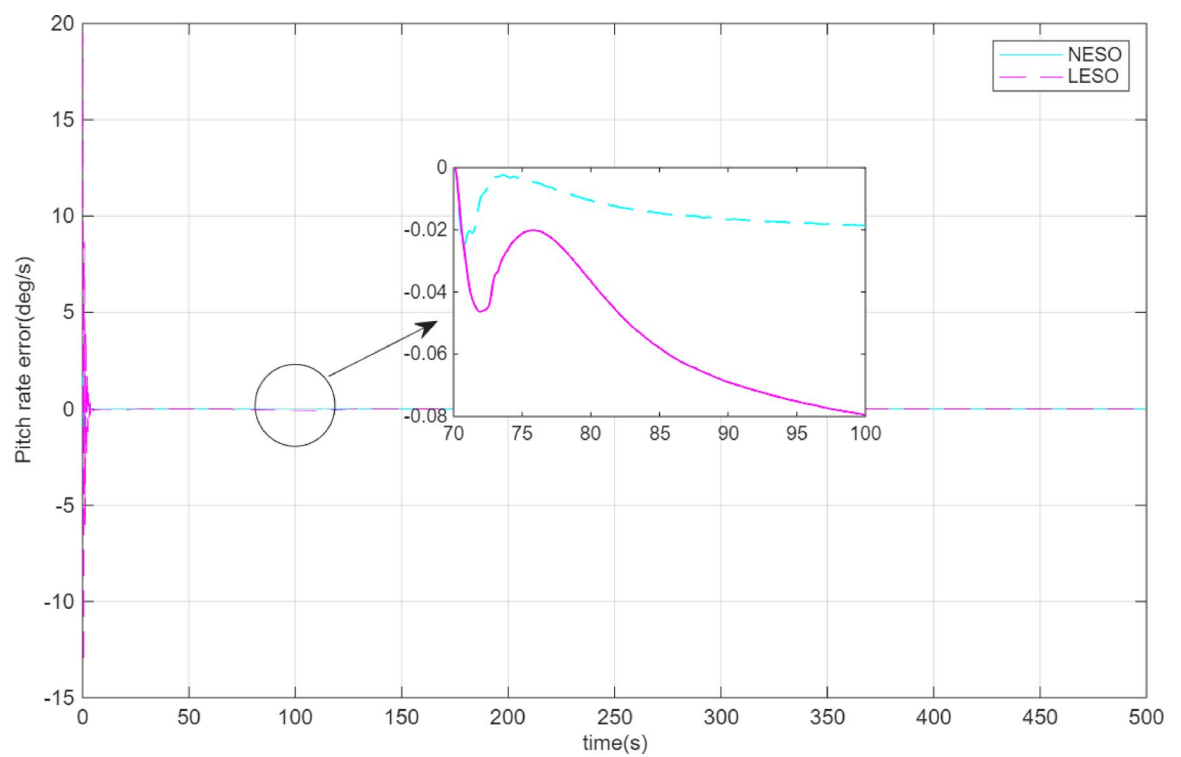


Fig. 24. Pitch rate angle estimation error.

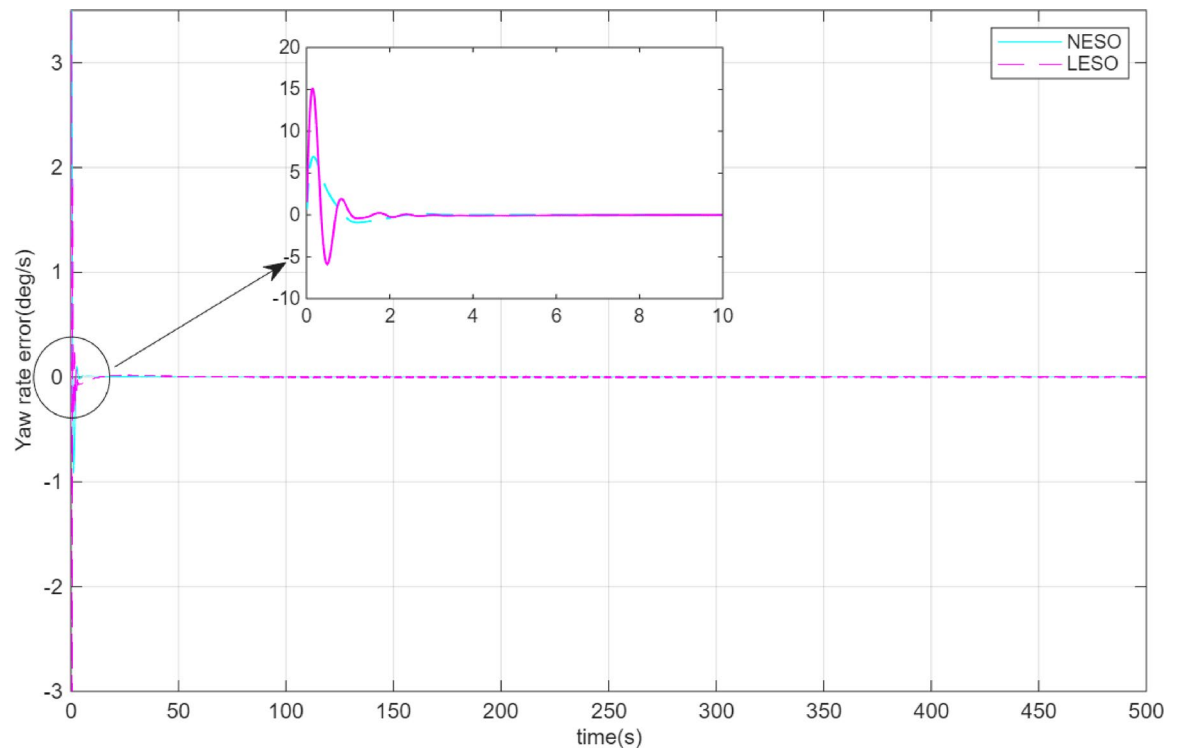


Fig. 25. Yaw rate angle estimation error.

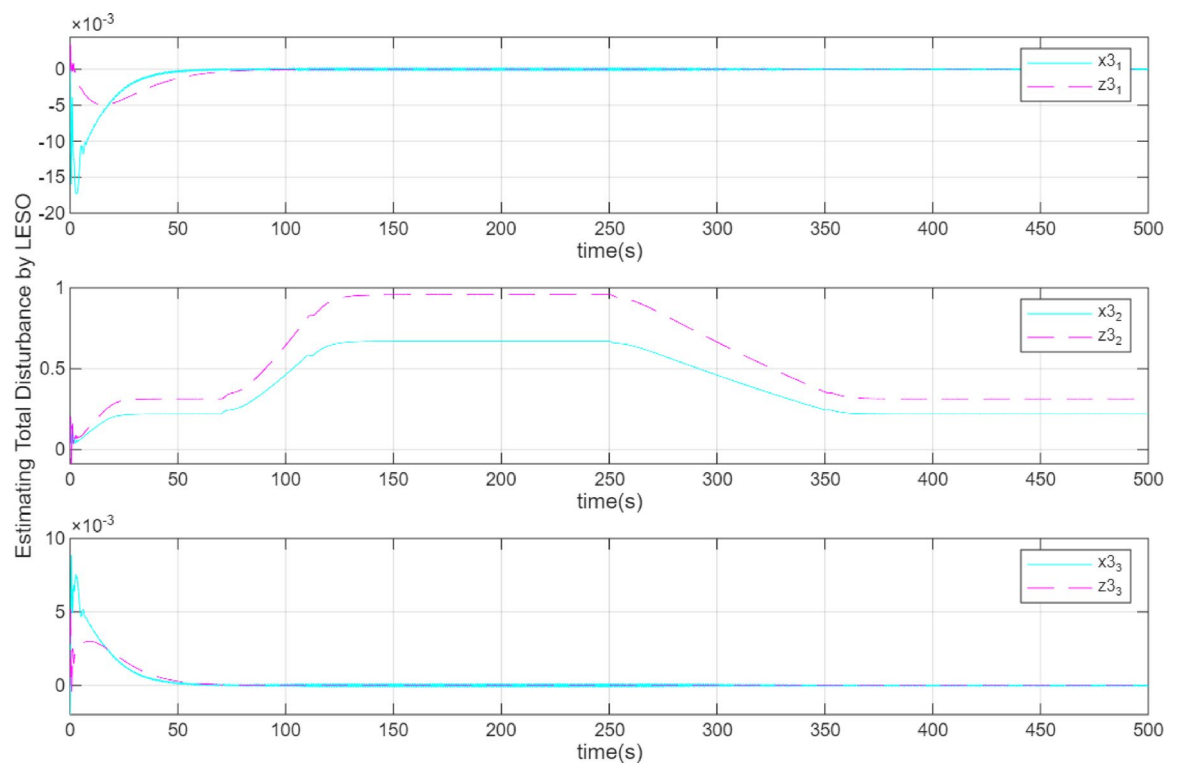


Fig. 26. Estimation of total disturbance by linear extended state observer.

Observer	NESO	LESO
Mean Squared Error	$mse_{\phi} = 1.164e-07$ $mse_{\theta} = 6.5777e-09$ $mse_{\psi} = 1.0954e-07$	$mse_{\phi} = 3.0054e-07$ $mse_{\theta} = 8.1889e-07$ $mse_{\psi} = 2.5109e-07$
Integral Absolute Error	$Iae_{\psi} = 0.0050291$ $Iae_{\theta} = 0.0040737$ $Iae_{\psi} = 0.0051942$	$Iae_{\psi} = 0.020702$ $Iae_{\theta} = 0.055711$ $Iae_{\psi} = 0.016357$
Integral Squared Error	$Ise_{\phi} = 5.1864e-05$ $Ise_{\theta} = 3.1321e-06$ $Ise_{\psi} = 4.8436e-05$	$Ise_{\phi} = 0.00014394$ $Ise_{\theta} = 0.00040311$ $Ise_{\psi} = 0.00011921$

Table 7. Comparison of linear and nonlinear extended state observers (LESO and NESO) under nominal conditions and in the presence of 20% uncertainty.

Observer	NESO	LESO
Mean Squared Error	$mse_{\phi} = 1.7182e-07$ $mse_{\theta} = 2.3907e-08$ $mse_{\psi} = 1.6528e-07$	$mse_{\phi} = 4.0456e-07$ $mse_{\theta} = 7.616e-07$ $mse_{\psi} = 3.7917e-07$
Integral Absolute Error	$Iae_{\phi} = 0.0058337$ $Iae_{\theta} = 0.0055213$ $Iae_{\psi} = 0.0062815$	$Iae_{\phi} = 0.015165$ $Iae_{\theta} = 0.055178$ $Iae_{\psi} = 0.01398$
Integral Squared Error	$Ise_{\phi} = 7.6097e-05$ $Ise_{\theta} = 1.0931e-05$ $Ise_{\psi} = 7.2831e-05$	$Ise_{\phi} = 0.00019247$ $Ise_{\theta} = 0.00037099$ $Ise_{\psi} = 0.00017978$

Table 8. Comparison of linear and nonlinear extended state observers (LESO and NESO) under external disturbance conditions.

Data availability

All data generated or analysed during this study are included in this published article.

Appendix A

A.1. Transfer Matrix

For accurate simulation of aircraft flight, it is essential to express all equations in the body frame. However, some equations of motion cannot be directly represented in the body frame and must be transformed from the wind frame. The forces and torques acting on the aircraft are typically defined in the wind frame, requiring a transformation to the body frame. The transfer matrix from the wind frame to the body frame is given as follows:

$$T_{WB} = \begin{bmatrix} \cos\alpha \cos\beta & \sin\beta & \sin\alpha \cos\beta \\ -\cos\alpha \sin\beta & \cos\beta & -\sin\alpha \sin\beta \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \quad (\text{A-1})$$

The transfer matrix between the wind frame and the body frame is expressed as follows:

$$T_{BW} = T_{WB}^T \quad (\text{A-2})$$

Aerodynamic forces in body frame :

$$\begin{bmatrix} F_{Ax} \\ F_{Ay} \\ F_{Az} \end{bmatrix}_B = T_{BW} \begin{bmatrix} F_{Ax} \\ F_{Ay} \\ F_{Az} \end{bmatrix}_w \quad (\text{A-3})$$

Aerodynamic moments in body frame :

$$\begin{bmatrix} L \\ M \\ N \end{bmatrix}_B = T_{BW} \begin{bmatrix} L \\ M \\ N \end{bmatrix}_w \quad (\text{A-4})$$

A.2. Linear Extended State Observer

The linear extended state observer for Eq. (12) is defined as follows:

$$\begin{cases} \dot{e} = z_1 - x_1 \\ \dot{z}_1 = z_2 - l_1 e \\ \dot{z}_2 = z_3 + b_0 u(t) - l_2 e \\ \dot{z}_3 = -l_3 e \end{cases} \quad (\text{A-5})$$

Where $z_1 = [\hat{\phi}, \hat{\theta}, \hat{\psi}]^T$ is the estimation vector of x_1 , $z_2 = [\hat{p}, \hat{q}, \hat{r}]^T$ is the estimation vector of x_2 , $z_3 = [\hat{H}_1(t), \hat{H}_2(t), \hat{H}_3(t)]^T$ is the extended vector used to estimate the total disturbance, $e = z_1 - x_1$ is the 3×1 vector of the angle estimation error, $l_i, i = 1, 2, 3$ is the 3×3 matrix of observer gains, and b_0 is the 3×3 matrix of control input gains.

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References

- Han, J. Auto-disturbances-rejection controller and its applications. *Control Decis.* **13** (1), 19–23 (1998).
- Gao, Z., Huang, Y. & Han, J. An alternative paradigm for control system design. *Proc. IEEE Conf. Decis. Control* **5**, 4578–4585. <https://doi.org/10.1109/CDC.2001.980926> (2001).
- Gao, Z. Scaling and Bandwidth-Parameterization based controller tuning. *Proc. Am. Control Conf.* **6**, 4989–4996. <https://doi.org/10.1109/acc.2003.1242516> (2003).
- Liu, H. & Li, S. Speed control for PMSM servo system using predictive functional control and extended state observer. *IEEE Trans. Ind. Electron.* **59** (2), 1171–1183. <https://doi.org/10.1109/TIE.2011.2162217> (2012).
- Sun, L., Li, D., Hu, K., Lee, K. Y. & Pan, F. On tuning and practical implementation of active disturbance rejection controller: A case study from a regenerative heater in a 1000 MW power plant. *Ind. Eng. Chem. Res.* **55** (23), 6686–6695. <https://doi.org/10.1021/acs.iecr.6b01249> (2016).
- Zheng, Q. On active disturbance rejection control: stability analysis and applications in disturbance decoupling control. *Dissertation no. July*, 1–102 (2009). <http://engagedscholarship.csuohio.edu/etdarchive/324/%0Apapers2://publication/uuid/59393457-288A-4839-8DF1-ECE06E975C36> [Online]. Available: .
- Ardalani, A., Jazi, S. H. & Malekzadeh, M. Implementation of Active Disturbance Rejection Control based on Super-Twisting Algorithm on Delta Robot, *10th RSI Int. Conf. Robot. Mechatronics, ICRoM 2022*, 539–545, (2022). <https://doi.org/10.1109/ICRoM57054.2022.10025271>
- Raouf, A. H., Yazdiniya, F. S. & Ansarifar, G. R. Super-twisting ADRC for maximum power point tracking control of photovoltaic power generation system based on non-linear extended state observer. *Heliyon* **10** (16), e36428. <https://doi.org/10.1016/j.heliyon.2024.e36428> (2024).
- Han, J. The extended state observer of a class of uncertain systems. *Control Decis.* **10** (1), 85–88 (1995).
- Huang, Y. & Han, J. Q. A new synthesis method for uncertain systems the Self-Stable region approach. *Int. J. Syst. Sci.* **30** (1), 33–38. <https://doi.org/10.1080/002077299292641> (1999).
- Hall, C. E. & Shtessel, Y. B. Sliding mode disturbance observer-based control for a reusable launch vehicle. *J. Guid Control Dyn.* **29** (6), 1315–1328. <https://doi.org/10.2514/1.20151> (2006).
- Mu, C., Sun, C., Qian, C. & Zhang, R. Super-twisting sliding mode control based on Lyapunov analysis for the cursing flight of hypersonic vehicles. *IEEE Int. Conf. Control Autom. ICCA.* (61125306), 522–527. <https://doi.org/10.1109/ICCA.2013.6565017> (2013).
- Wang, Y., Fang, S., Hu, J. & Huang, D. A novel active disturbance rejection control of PMSM based on deep reinforcement learning for more electric aircraft. *IEEE Trans. Energy Convers.* **38** (2), 1461–1470. <https://doi.org/10.1109/TEC.2023.3235927> (2023).
- Du, C. et al. Research on active disturbance rejection control with parameter autotune mechanism for induction motors based on adaptive particle swarm optimization algorithm with dynamic inertia weight. *IEEE Trans. Power Electron.* **34** (3), 2841–2855. <https://doi.org/10.1109/TPEL.2018.2841869> (2019).
- Roskam, J. *Airplane Flight Dynamics and Automatic Flight Controls Pt. 1. Darcorporation.* (1998).
- Durham, W. *Aircraft flight dynamics and control.* (2008).
- Chiel, H. J. & Thomas, P. J. Special issue on applied neurodynamics: From neural dynamics to neural engineering, **8**(6). (2011). <https://doi.org/10.1088/1741-2552/8/6/060201>
- Derafa, L., Benallegue, A. & Fridman, L. Super twisting control algorithm for the attitude tracking of a four rotors UAV. *J. Frankl. Inst.* **349** (2), 685–699. <https://doi.org/10.1016/j.jfranklin.2011.10.011> (2012).
- Yan, L., Chen, X., Chen, Y. & Wen, J. A cooperative charging control strategy for electric vehicles based on multiagent deep reinforcement learning. *IEEE Trans. Ind. Inf.* **18** (12), 8765–8775. <https://doi.org/10.1109/TII.2022.3152218> (2022).
- Farebrother, J., Machado, M. C. & Bowling, M. Generalization and Regularization in DQN, 1–29, [Online]. (2018). Available: <http://arxiv.org/abs/1810.00123>
- Hou, Y., Liu, L., Wei, Q., Xu, X. & Chen, C. A novel DDPG method with prioritized experience replay, *2017 IEEE Int. Conf. Syst. Man, Cybern. SMC 2017*, **2017**, 316–321, (2017). <https://doi.org/10.1109/SMC.2017.8122622>
- Roskam, J. *Airplane Flight Dynamics and Automatic Flight Controls - Part I.* (1998).

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Credit author statementFatemehSadat Yazdiniya: Conceptualization, Methodology, Investigation, Software, Validation, Visualization, Writing- Original draft preparation, Reviewing and Editing.GholamReza Ansarifar: Supervision, Methodology, Validation, Writing- Reviewing and Editing, Resources.

Declarations

Competing interests

The authors declare no competing interests.

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